



Statistical Fine-Mapping: From GWAS Signals to Causal Variants

TUTORIAL

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0.0.1 What is Fine-Mapping?

Genome-wide association studies (GWAS) have transformed human genetics by identifying thousands of genomic regions associated with diseases and complex traits. However, GWAS typically does not identify the exact causal variant responsible for the observed association.

Instead, GWAS identifies a region of the genome containing many correlated variants.

This happens because of **Linkage Disequilibrium (LD)**, the non-random correlation between nearby genetic variants.

For example:

SNP A ---- SNP B ---- SNP C
 ^
 causal

If SNP B is causal, SNP A and SNP C may also appear significant because they are inherited together.

Therefore:

GWAS answers:

Which genomic region is associated with the trait?

Fine-mapping answers:

Which specific variant is most likely causing the association?

0.0.2 Why GWAS Alone Is Not Enough

Suppose a GWAS identifies a significant locus on chromosome 22.

The Manhattan plot may show a strong peak:

```

      *
     ***
    *****
   *********

```

The peak could contain:

- 10 SNPs
- 100 SNPs
- 1000 SNPs

all showing nearly identical p-values.

GWAS cannot determine which SNP is truly causal because many neighboring SNPs are correlated through LD.

For example:

```

rs1
rs2
rs3
rs4
rs5

```

All five SNPs may have nearly identical p-values.

Fine-mapping attempts to separate these signals and determine which SNP has the strongest evidence for causality.

0.0.3 What Is the Goal of Fine-Mapping?

Fine-mapping attempts to estimate:

$P(\text{SNP is causal} \mid \text{observed data})$

for every SNP in a genomic region.

Instead of reporting only p-values, fine-mapping reports probabilities.

These probabilities are called:

0.0.3.1 Posterior Inclusion Probabilities (PIPs)

For example:

```
rs1 : 0.01  
rs2 : 0.04  
rs3 : 0.88  
rs4 : 0.05  
rs5 : 0.02
```

Interpretation:

```
rs3 has an 88% probability of being causal
```

and is therefore the strongest candidate variant.

0.0.4 GWAS Versus Fine-Mapping

GWAS is designed to detect associated regions.

Fine-mapping is designed to identify likely causal variants within those regions.

GWAS focuses on:

- p-values
- association testing
- genome-wide discovery

Fine-mapping focuses on:

- posterior probabilities
- causal inference
- variant prioritization

In modern statistical genetics, GWAS is often considered the first step, while fine-mapping is the second step.

0.0.5 Why Fine-Mapping Matters

Imagine a schizophrenia GWAS identifies a significant locus containing 200 SNPs.

Experimental validation of 200 variants would be extremely expensive.

Fine-mapping may reduce the candidate list to:

3 SNPs

or even

1 SNP

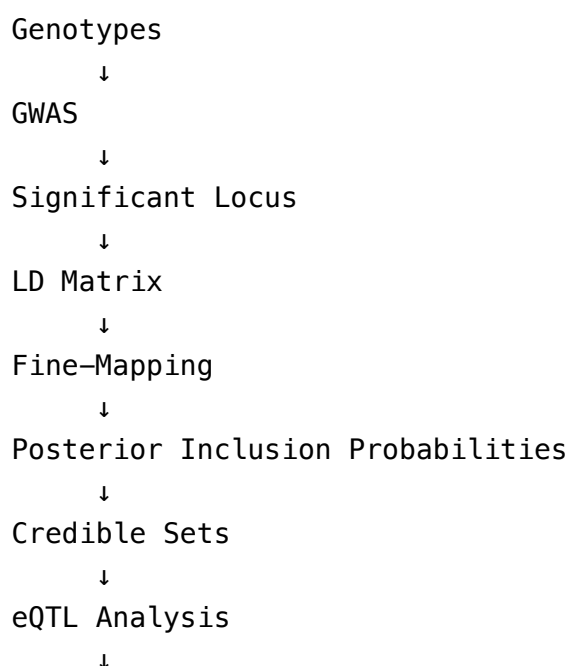
This dramatically reduces the cost and complexity of downstream biological experiments.

Fine-mapping is therefore essential for:

- Functional genomics
- Drug target discovery
- eQTL interpretation
- Colocalization analyses
- Precision medicine

0.0.6 The Modern Statistical Genetics Pipeline

Most modern studies follow a workflow similar to:



Colocalization



Candidate Gene

This pipeline forms the backbone of many large-scale genetics projects including:

- UK Biobank
- FinnGen
- Psychiatric Genomics Consortium
- GTEx

0.0.7 What Information Does Fine-Mapping Need?

The first question in any fine-mapping study is:

What type of data is available?

There are three common situations.

0.0.7.1 Scenario 1: Individual-Level Data

You have access to:

Genotypes
Phenotypes

for every participant.

This is the ideal scenario because LD can be computed directly from the study samples.

0.0.7.2 Scenario 2: Single-Cohort GWAS Summary Statistics

You only have:

GWAS summary statistics

such as:

- beta
- standard error

- p-value

but no individual-level genotypes.

In this case, an external LD reference panel is required.

Examples include:

- 1000 Genomes
 - UK Biobank
 - TOPMed
-

0.0.7.3 Scenario 3: Meta-Analysis GWAS

You only have meta-analysis summary statistics from multiple cohorts.

This is the most challenging situation because LD differs across cohorts.

Specialized methods such as:

- FastMap
- CARMA
- SLALOM

are often required.

0.0.8 The Central Role of Linkage Disequilibrium

Linkage Disequilibrium (LD) is the most important concept in fine-mapping.

Suppose two SNPs have:

$$r^2 = 0.95$$

They are almost perfectly correlated.

Statistically, it becomes difficult to distinguish which SNP is causal.

High LD leads to:

Large credible sets
Low resolution

Low LD leads to:

Small credible sets

High resolution

This is why ancestry-matched LD reference panels are extremely important.

0.0.9 A Simple Numerical Example

Consider five SNPs:

```
import pandas as pd

results = pd.DataFrame({
    "SNP": ["rs1", "rs2", "rs3", "rs4", "rs5"],
    "PIP": [0.01, 0.04, 0.88, 0.05, 0.02]
})

results
```

Output:

```
rs1  0.01
rs2  0.04
rs3  0.88
rs4  0.05
rs5  0.02
```

The most likely causal SNP is:

rs3

because it has the highest Posterior Inclusion Probability.

0.0.10 Key Concepts Introduced in This Tutorial

Before learning specific fine-mapping methods, you should understand:

- Linkage Disequilibrium (LD)
- Causal Variants
- Posterior Inclusion Probability (PIP)
- Credible Sets

- LD Reference Panels
- Summary Statistics
- Individual-Level Data

These concepts form the foundation of all modern fine-mapping approaches.

0.0.11 Key Takeaways

Fine-mapping is the process of identifying likely causal variants within a GWAS locus. Because neighboring SNPs are often correlated through linkage disequilibrium, GWAS alone cannot determine which variant is responsible for an association signal. Fine-mapping combines GWAS statistics with LD information and Bayesian inference to assign probabilities of causality to variants. The primary outputs are Posterior Inclusion Probabilities (PIPs) and Credible Sets. Fine-mapping serves as the bridge between GWAS discovery and downstream analyses such as eQTL mapping, colocalization, functional validation, and drug target identification.

0.0.12 Part 2: Linkage Disequilibrium, Posterior Inclusion Probabilities (PIPs), and Credible Sets

Before we can understand methods such as ABF, SuSiE, FINEMAP, CARMA, or FastMap, we must understand three foundational concepts:

1. Linkage Disequilibrium (LD)
2. Posterior Inclusion Probabilities (PIPs)
3. Credible Sets

These concepts form the mathematical backbone of statistical fine-mapping.

0.0.13 What Is Linkage Disequilibrium (LD)?

Linkage Disequilibrium (LD) measures the correlation between genetic variants.

When two SNPs are inherited together more often than expected by chance, they are said to be in LD.

For example:

SNP1 ----- SNP2 ----- SNP3

Suppose:

$$r^2(\text{SNP1}, \text{SNP2}) = 0.95$$

This means:

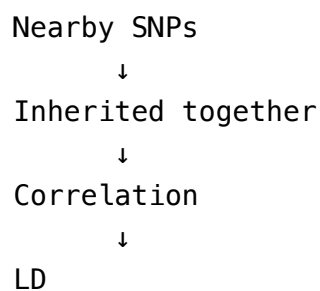
- Knowing SNP1 almost perfectly predicts SNP2.
 - Their genotypes are highly correlated.
 - They tend to be inherited together.
-

0.0.14 Why Does LD Exist?

During reproduction, chromosomes undergo recombination.

However, nearby variants are less likely to be separated.

Therefore:



The closer two SNPs are physically located, the stronger their LD tends to be.

0.0.15 LD Measures

Two common measures are:

0.1 D'

Measures historical recombination.

Range:

$$0 \rightarrow 1$$

0.2 r^2

Measures correlation.

Range:

$$0 \rightarrow 1$$

Interpretation:

$$r^2 = 0$$

No correlation

$$r^2 = 1$$

Perfect correlation

Fine-mapping primarily uses:

$$r^2$$

0.2.1 Example LD Matrix

Consider four SNPs.

```
import pandas as pd

ld = pd.DataFrame(
    [
        [1.00, 0.92, 0.15, 0.05],
        [0.92, 1.00, 0.18, 0.07],
        [0.15, 0.18, 1.00, 0.85],
        [0.05, 0.07, 0.85, 1.00]
    ],
    columns=["rs1", "rs2", "rs3", "rs4"],
    index=["rs1", "rs2", "rs3", "rs4"]
)

ld
```

Output:

	rs1	rs2	rs3	rs4
rs1	1.00	0.92	0.15	0.05
rs2	0.92	1.00	0.18	0.07
rs3	0.15	0.18	1.00	0.85
rs4	0.05	0.07	0.85	1.00

Interpretation:

rs1 and rs2 form one LD block
rs3 and rs4 form another LD block

0.2.2 Why LD Causes Problems in GWAS

Suppose rs1 is truly causal.

Because rs2 has:

$$r^2 = 0.92$$

both SNPs may show significant p-values.

Example:

rs1 p = 1e-10
rs2 p = 2e-10

GWAS cannot distinguish:

Which SNP is causal?

Both appear significant.

This is the central problem that fine-mapping attempts to solve.

0.2.3 The Fine-Mapping Perspective

Instead of asking:

Which SNP has the smallest p-value?

Fine-mapping asks:

Which SNP is most likely causal?

This distinction is extremely important.

A SNP with the smallest p-value is not always the causal SNP.

0.2.4 Posterior Inclusion Probability (PIP)

The primary output of modern fine-mapping methods is the:

0.2.4.1 Posterior Inclusion Probability

Mathematically:

$$\text{PIP} = P(\text{SNP is causal} \mid \text{data})$$

PIP values range from:

$$0 \rightarrow 1$$

Interpretation:

0
Not likely causal

1
Almost certainly causal

0.2.5 Example PIPs

Suppose fine-mapping returns:

```
import pandas as pd

pip = pd.DataFrame({
    "SNP": ["rs1", "rs2", "rs3", "rs4"],
    "PIP": [0.03, 0.07, 0.82, 0.08]
})
```

```
pip
```

Output:

```
rs1  0.03
rs2  0.07
rs3  0.82
rs4  0.08
```

Interpretation:

```
rs3 has an 82% probability
of being causal
```

0.2.6 PIPs Are Not p-values

Many beginners confuse PIPs with p-values.

They are completely different.

0.2.6.0.1 p-value

Measures:

```
Evidence against the null hypothesis
```

0.2.6.0.2 PIP

Measures:

```
Probability of causality
```

Example:

```
p-value = 1e-20
```

does NOT mean

```
99.99999999999999% chance causal
```

That interpretation is incorrect.

0.2.7 Multiple Causal Variants

Many loci contain more than one causal SNP.

Example:

Variant 20

Variant 39

Variant 68

All may influence the same trait.

This is exactly why methods such as SuSiE were developed. The workshop practical simulates three causal variants in the same locus and shows that a locus may contain multiple independent signals.

0.2.8 What Is a Credible Set?

A Credible Set is a group of variants that collectively contain the causal SNP with high probability.

Most commonly:

95% Credible Set

Meaning:

There is a 95% probability
that the causal SNP is inside
this set.

0.2.9 Example Credible Set

Suppose:

rs1 PIP = 0.60

rs2 PIP = 0.25

rs3 PIP = 0.08

rs4 PIP = 0.04

rs5 PIP = 0.03

Cumulative probabilities:

$$rs1 = 0.60$$

$$rs1 + rs2 = 0.85$$

$$rs1 + rs2 + rs3 = 0.93$$

$$rs1 + rs2 + rs3 + rs4 = 0.97$$

The 95% credible set becomes:

rs1

rs2

rs3

rs4

because together they exceed 95%.

0.2.10 Small Versus Large Credible Sets

Small credible set:

rs25

Excellent resolution.

Medium credible set:

rs25

rs26

rs27

Reasonable resolution.

Large credible set:

50 SNPs

Poor resolution.

0.2.11 What Determines Credible Set Size?

Several factors influence fine-mapping resolution.

0.2.11.1 Sample Size

Larger sample sizes:

Higher power

Smaller credible sets

The workshop demonstrates that increasing sample size improves PIPs and fine-mapping resolution.

0.2.11.2 LD Structure

High LD:

Large credible sets

Low LD:

Small credible sets

The workshop also shows that increasing LD makes causal variants harder to resolve.

0.2.11.3 Effect Size

Larger effects:

Higher PIPs

Smaller effects:

Lower PIPs

0.2.12 Visual Interpretation

Ideal situation:

rs39
PIP = 0.98

Credible Set:

{rs39}

Excellent.

Difficult situation:

rs39 = 0.20
rs40 = 0.19
rs41 = 0.18
rs42 = 0.17
rs43 = 0.16

Credible Set:

5 variants

Less certainty.

0.2.13 Why Fine-Mapping Uses Bayesian Statistics

Bayesian methods naturally answer:

How likely is each SNP
to be causal?

which is exactly the scientific question we care about.

This is why nearly all modern fine-mapping methods:

- ABF
- SuSiE
- FINEMAP
- CARMA
- CAVIAR

are Bayesian methods.

0.2.14 Key Takeaways

Linkage Disequilibrium is the correlation structure among nearby variants and is the primary reason GWAS cannot identify causal variants directly. Fine-mapping uses LD information together with GWAS statistics to estimate Posterior Inclusion Probabilities (PIPs), which represent the probability that a variant is causal. Variants with high PIPs are prioritized for follow-up studies. Because uncertainty remains, fine-mapping methods typically report Credible Sets, collections of variants that together contain the causal variant with high probability. Sample size, effect size, and LD structure strongly influence PIPs and credible set sizes.

0.2.15 Part 3: Bayesian Fine-Mapping and Approximate Bayes Factors (ABF)

In the previous section, we learned that GWAS identifies associated regions but cannot reliably determine which variant is causal because of Linkage Disequilibrium (LD).

The next question becomes:

How can we assign probabilities of causality to SNPs?

One of the earliest and most influential solutions is:

0.2.16 Approximate Bayes Factors (ABF)

ABF forms the foundation of many modern fine-mapping methods and remains widely used today.

The ISG workshop uses Jon Wakefield's Approximate Bayes Factor approach before introducing SuSiE.

0.2.17 Why Move Beyond p-values?

Suppose a GWAS reports:

rs1	$p = 1 \times 10^{-8}$
rs2	$p = 2 \times 10^{-8}$
rs3	$p = 5 \times 10^{-8}$

All three SNPs are significant.

But p-values do not tell us:

Which SNP is causal?

They only tell us:

How unlikely the data would be
if no association existed.

Fine-mapping requires something different:

Probability that a SNP is causal.

This is where Bayesian statistics becomes useful.

0.2.18 Frequentist Versus Bayesian Thinking

0.2.19 Frequentist Approach

The null hypothesis is:

$$\beta = 0$$

We calculate:

$$P(\text{Data} \mid \text{Null})$$

which produces a p-value.

0.2.20 Bayesian Approach

We compare:

Model 1:
SNP is causal

versus

Model 0:
 SNP is not causal

and calculate:

$$P(\text{Causal} \mid \text{Data})$$

This is exactly the quantity we want.

0.2.21 What Is a Bayes Factor?

A Bayes Factor compares two competing models.

Mathematically:

$$\text{BF} = \frac{P(\text{Data} \mid \text{Causal Model})}{P(\text{Data} \mid \text{Null Model})}$$

Interpretation:

BF = 1
 No preference

BF = 10
 Data are 10 times more likely
 under the causal model

BF = 100
 Strong evidence for causality

0.2.22 Why “Approximate” Bayes Factor?

Computing exact Bayes Factors can be computationally expensive.

Wakefield proposed an approximation that requires only:

- Effect size estimate (β)
- Standard error (SE)

These quantities are available in almost every GWAS summary statistics file.

Therefore:

```
Summary Statistics
    ↓
ABF
    ↓
Posterior Probability
```

No individual-level data are required.

0.2.23 Inputs Required for ABF

For each SNP we need:

Beta

Example:

$$\beta = 0.12$$

and

$$SE = 0.03$$

From these values we compute:

$$Z = \beta / SE$$

0.2.24 Example

Suppose:

```
beta = 0.12
se = 0.03

z = beta / se

print(z)
```

Output:

4.0

A large absolute Z-score indicates stronger evidence for association.

0.2.25 Wakefield's Approximate Bayes Factor

The workshop implements the following ABF calculation.

First define:

$$V = SE^2$$

Then choose a prior variance:

W

A common value:

$$W = 0.04$$

Next calculate:

$$r = W / (W + V)$$

The log Bayes Factor becomes:

$$\ln BF = (\log(1-r) + rZ^2)/2$$

This transforms GWAS statistics into Bayesian evidence.

0.2.26 Understanding the Prior Variance W

The parameter:

W

represents our prior belief about effect sizes.

Small W :

Expect small effects

Large W :

Allow larger effects

Typical fine-mapping analyses use values between:

0.01 and 0.04

0.2.27 From Bayes Factors to Posterior Probabilities

After computing Bayes Factors for every SNP:

```
rs1    BF = 3
rs2    BF = 50
rs3    BF = 10
```

we normalize them.

Suppose:

Total BF = 63

Then:

```
PIP(rs1) = 3/63 = 0.048
PIP(rs2) = 50/63 = 0.794
PIP(rs3) = 10/63 = 0.159
```

These are the Posterior Inclusion Probabilities.

0.2.28 A Complete ABF Example

```
import numpy as np
import pandas as pd
```

```

beta = np.array([0.05,0.15,0.09])
se = np.array([0.03,0.03,0.03])

W = 0.04

z = beta/se
V = se**2

r = W/(W+V)

lbf = (np.log(1-r) + r*(z**2))/2

bf = np.exp(lbf)

pip = bf / bf.sum()

results = pd.DataFrame({
    "Beta":beta,
    "SE":se,
    "Z":z,
    "BF":bf,
    "PIP":pip
})

results

```

0.2.29 Interpreting the Results

Suppose the output is:

```

SNP1  PIP = 0.05
SNP2  PIP = 0.82
SNP3  PIP = 0.13

```

Interpretation:

```

SNP2 is the strongest
causal candidate.

```

0.2.30 The Single-Causal-Variant Assumption

ABF makes a very important assumption:

Only one causal variant
exists in the region.

This assumption simplifies the mathematics.

However, many real loci contain:

2
3
5
or more
causal variants

This limitation motivated the development of more advanced methods such as:

- SuSiE
- FINEMAP
- CAVIAR
- CARMA

0.2.31 Why ABF Sometimes Fails

Imagine:

Variant 20 is causal
Variant 39 is causal
Variant 68 is causal

ABF assumes:

Only one can be causal.

Therefore:

Signal splitting occurs

and the true signals may receive lower PIPs.

The workshop demonstrates exactly this scenario by simulating three causal variants in the same locus.

0.2.32 Advantages of ABF

ABF is still popular because it is:

- Fast
- Easy to compute
- Requires only summary statistics
- Does not require an LD matrix
- Computationally efficient

This makes it valuable for large GWAS datasets.

0.2.33 Limitations of ABF

ABF assumes:

`One causal variant per locus`

which is often unrealistic.

Other limitations:

- Cannot separate multiple signals
- Can be confused by strong LD
- Lower resolution in complex regions

These limitations motivated modern methods such as SuSiE.

0.2.34 ABF in Colocalization

ABF is also used in:

`coloc.abf`

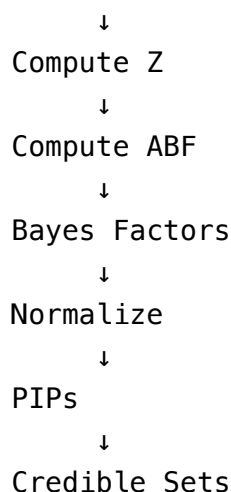
for testing whether GWAS and eQTL signals share the same causal variant.

The colocalization workshop specifically notes that `coloc.abf` requires only summary statistics and assumes at most one causal variant per locus.

This makes ABF an important building block in statistical genetics.

0.2.35 Visual Summary

GWAS Summary Statistics



0.2.36 Key Takeaways

Approximate Bayes Factors (ABF) were among the first practical Bayesian fine-mapping methods. They use GWAS summary statistics to estimate evidence that each SNP is causal. By normalizing Bayes Factors across SNPs, ABF produces Posterior Inclusion Probabilities (PIPs), which quantify the probability that a variant is causal. ABF is computationally efficient and requires only summary statistics, making it useful when LD information is unavailable. However, its major limitation is the assumption that only one causal variant exists within a locus. Because many real loci contain multiple independent signals, more advanced methods such as SuSiE were developed to overcome this limitation.

0.2.37 Part 4: SuSiE (Sum of Single Effects)

Approximate Bayes Factors (ABF) introduced an important idea:

```
Estimate the probability  
that each SNP is causal.
```

However, ABF assumes:

```
One causal variant  
per locus
```

Unfortunately, real biological loci rarely behave this way.

Many GWAS regions contain:

2
3
5
or more
independent causal variants

To solve this problem, modern fine-mapping methods such as:

- SuSiE
- FINEMAP
- CARMA

were developed.

Among these, SuSiE has become one of the most widely used methods in statistical genetics.

The ISG workshop uses SuSiE as the primary modern fine-mapping method.

0.2.38 What Does SuSiE Mean?

SuSiE stands for:

0.2.38.1 Sum of Single Effects

The key idea is surprisingly simple.

Instead of assuming:

One causal SNP

SuSiE assumes:

Several causal SNPs

and models the phenotype as the sum of multiple independent effects.

Mathematically:

Phenotype

=

Effect 1

+
 Effect 2
 +
 Effect 3
 +
 ...

Hence:

Sum of Single Effects

0.2.39 Why Was SuSiE Developed?

Imagine a locus containing:

Variant 20
 Variant 39
 Variant 68

Suppose all three are truly causal.

ABF attempts to explain the locus using:

One SNP

SuSiE attempts to explain the locus using:

Multiple SNPs

This is much closer to biological reality.

The workshop simulation deliberately contains three causal variants in the same region.

0.2.40 The Core SuSiE Model

SuSiE models the phenotype as:

$$y = Xb + e$$

where:

y

is the phenotype,

X

is the genotype matrix,

b

contains SNP effects,

and

e

is random noise.

0.2.41 The Single-Effect Idea

Instead of fitting all SNP effects simultaneously, SuSiE decomposes the model into:

Effect 1
Effect 2
Effect 3
...

Each effect identifies one potential causal signal.

For example:

Signal 1 → Variant 39

Signal 2 → Variant 20

Signal 3 → Variant 68

The final model combines them.

0.2.42 The Parameter L

A very important SuSiE parameter is:

L

which specifies the maximum number of causal signals.

Example:

L = 4

means:

Allow up to
4 independent signals

The workshop uses:

```
allowed_causal <- L + 1
```

and then fits:

```
susie_rss(  
  ...  
  L = allowed_causal  
)
```

to search for multiple independent signals.

0.2.43 What Does SuSiE Produce?

SuSiE returns two important outputs:

0.2.43.1 Posterior Inclusion Probabilities

PIP

for every SNP.

0.2.43.2 Credible Sets

Groups of SNPs likely to contain causal variants.

Example:

CS1:
rs39

CS2:
rs20

Each credible set corresponds to a distinct signal.

0.2.44 Example: Three Causal Variants

Suppose the truth is:

Variant 20
Variant 39
Variant 68

ABF may identify:

Variant 39 only

because it assumes one signal.

SuSiE can identify:

Signal 1 → Variant 20

Signal 2 → Variant 39

Signal 3 → Variant 68

This is the major advantage of SuSiE.

0.2.45 Running SuSiE in R

The workshop uses:


```
library(susieR)

susie_results <- susie_rss(
  bhat = GWAS_df$beta_marginal,
  shat = GWAS_df$se_marginal,
  n = N,
  R = in_sample_LD,
  var_y = var(y),
  L = 4,
  estimate_residual_variance = TRUE
)
```

Notice the inputs:

- Beta estimates
- Standard errors
- Sample size
- LD matrix
- Phenotype variance

Unlike ABF:

SuSiE requires LD

because it explicitly models multiple correlated signals.

0.2.46 Why Does SuSiE Need an LD Matrix?

Suppose:

rs39

and

rs40

have:

$$r^2 = 0.95$$

SuSiE needs LD information to determine whether:

One signal

or

Two independent signals

exist.

Without LD, this distinction becomes impossible.

0.2.47 Posterior Inclusion Probabilities in SuSiE

Like ABF, SuSiE reports PIPs.

Example:

rs20 0.95

rs39 0.98

rs68 0.92

Interpretation:

All three SNPs
are likely causal.

This is much more informative than a single p-value.

0.2.48 Credible Sets in SuSiE

Suppose SuSiE reports:

CS1 = {rs20}

CS2 = {rs39}

CS3 = {rs68}

This is excellent resolution.

More commonly:

```
CS1 =  
rs20  
rs21  
rs22
```

meaning:

```
One of these SNPs  
is probably causal
```

but the data cannot distinguish them perfectly.

0.2.49 Visualizing SuSiE Results

The workshop visualizes:

```
susie_plot(  
  susie_results,  
  y = "PIP"  
)
```

Each point represents a SNP.

The y-axis shows:

```
Posterior Inclusion Probability
```

High peaks indicate likely causal variants.

0.2.50 When Does SuSiE Work Best?

SuSiE performs best when:

0.2.51 Large sample size

```
More information
```

0.2.52 Lower LD

Better separation
of causal variants

0.2.53 Strong effects

Higher PIPs

0.2.54 Accurate LD matrix

Correct ancestry matching

0.2.55 Common Mistake

Many researchers assume:

The lead SNP
is causal.

SuSiE often demonstrates that:

Lead SNP ≠ Causal SNP

because the association signal may be distributed across correlated variants.

0.2.56 SuSiE Versus ABF

Feature	ABF	SuSiE
Bayesian	Yes	Yes
Uses Summary Statistics	Yes	Yes
Uses LD Matrix	No	Yes
Multiple Signals	No	Yes
Credible Sets	Limited	Yes
Modern Standard	No	Yes

0.2.57 Why SuSiE Became Popular

SuSiE provides:

- Accurate PIPs
- Multiple signal detection
- Credible sets
- Computational efficiency
- Summary-statistics support
- Compatibility with colocalization

For these reasons, it has become one of the most widely used fine-mapping methods in modern GWAS studies.

0.2.58 Key Takeaways

SuSiE (Sum of Single Effects) is a Bayesian fine-mapping method designed to identify multiple causal variants within a genomic region. Unlike ABF, which assumes only one causal SNP per locus, SuSiE models several independent signals simultaneously. It requires GWAS summary statistics together with an LD matrix and produces Posterior Inclusion Probabilities (PIPs) and Credible Sets for each signal. SuSiE is currently one of the most popular fine-mapping methods because it provides accurate causal variant prioritization while accommodating the complexity of real genetic loci.

0.2.59 Part 5: Running a Complete Fine-Mapping Analysis Step-by-Step

So far we have learned:

- What fine-mapping is
- Why GWAS cannot identify causal variants
- How ABF works
- How SuSiE improves upon ABF

Now it is time to walk through a complete fine-mapping workflow similar to the ISG workshop practical. We will:

```
Simulate Genotypes
  ↓
Simulate Phenotype
  ↓
```

```
Run GWAS
  ↓
Run ABF
  ↓
Run SuSiE
  ↓
Interpret PIPs
  ↓
Interpret Credible Sets
```

This is essentially the workflow followed in many modern GWAS fine-mapping studies.

0.2.60 Step 1: Import Required Packages

```
library(MASS)
library(susieR)
library(ggplot2)
```

Package roles:

MASS	→ Data simulation
susieR	→ Fine-mapping
ggplot2	→ Visualization

0.2.61 Step 2: Simulate a Genetic Region

To understand fine-mapping, we first create a synthetic genomic region where we know the true causal variants.

Suppose:

```
p <- 100
```

meaning:

100 SNPs

in our locus.

0.2.62 Create an LD Matrix

The workshop simulates a region where all SNPs have moderate correlation.

```
p <- 100

LD <- matrix(
  0.3,
  nrow = p,
  ncol = p
)

diag(LD) <- 1
```

This produces:

```
Off-diagonal = 0.3
Diagonal = 1
```

representing moderate LD.

0.2.63 Visualize the LD Matrix

```
ld_df <- expand.grid(
  SNP1 = 1:p,
  SNP2 = 1:p
)

ld_df$LD <- as.vector(LD)

ggplot(
  ld_df,
  aes(
    SNP1,
    SNP2,
    fill = LD
  )
) +
geom_raster() +
coord_fixed()
```

Interpretation:

Red = Strong LD

White = Weak LD

0.2.64 Step 3: Simulate Genotypes

Now generate genotypes for:

```
N <- 1000
```

individuals.

```
set.seed(4)

X <- mvrnorm(
  n = N,
  mu = rep(0,p),
  Sigma = LD
)
```

Dimensions:

```
dim(X)
```

Output:

```
1000 × 100
```

Meaning:

```
1000 individuals
100 SNPs
```

0.2.65 Step 4: Define Causal Variants

Suppose:

```
L <- 3
```

meaning:

```
3 true causal SNPs
```


Select them randomly.

```
set.seed(1)

causal_ind <- sample(
  1:p,
  L,
  replace = FALSE
)

causal_ind
```

Example output:

```
68
39
1
```

These are the ground-truth causal variants.

The workshop uses this exact setup.

0.2.66 Step 5: Assign Effect Sizes

Assume:

```
h2g <- 0.1
```

or:

10% SNP heritability

Assign effects.

```
per_snp_h2g <- h2g / L

effect_sizes <- rnorm(
  L,
  mean = 0,
  sd = sqrt(per_snp_h2g)
)
```

Example:

Variant 68 → 0.18

Variant 39 → -0.23

Variant 1 → 0.14

0.2.67 Step 6: Construct the Genetic Effect

Create beta coefficients.

```
beta <- rep(0,p)

beta[causal_ind] <- effect_sizes
```

Calculate the genetic component.

```
genetic_effect <- X %*% beta
```

0.2.68 Step 7: Simulate Phenotypes

Add environmental noise.

```
var_g <- var(genetic_effect)

sigma_squared <- ifelse(
  1-var_g > 0.05,
  1-var_g,
  0.1
)

epsilon <- rnorm(
  N,
  mean = 0,
  sd = sqrt(sigma_squared)
)

y <- as.numeric(
  genetic_effect + epsilon
)
```

Now we have:

Genotypes

Phenotype

and can run GWAS.

0.2.69 Step 8: Run GWAS

Create a results table.

```
GWAS_df <- data.frame(  
  beta = numeric(p),  
  se = numeric(p),  
  z = numeric(p),  
  pval = numeric(p)  
)
```

Loop through SNPs.

```
for(i in 1:p){  
  
  fit <- summary(  
    lm(y ~ X[,i] - 1)  
  )  
  
  beta_hat <- fit$coefficients[  
    1,  
    "Estimate"  
  ]  
  
  se_hat <- fit$coefficients[  
    1,  
    "Std. Error"  
  ]  
  
  z <- beta_hat / se_hat  
  
  p <- pchisq(  
    z^2,  
    df = 1,  
    lower.tail = FALSE  
  )  
  
  GWAS_df[i,] <- c(  
    beta_hat,
```

```
    se_hat,  
    z,  
    p  
  )  
}
```

This performs:

100 separate GWAS tests

0.2.70 Step 9: Visualize GWAS Results

```
GWAS_df$SNP <- 1:nrow(GWAS_df)  
  
GWAS_df$minus_log10p <-  
  -log10(GWAS_df$pval)  
  
ggplot(  
  GWAS_df,  
  aes(  
    SNP,  
    minus_log10p  
  )  
) +  
  geom_point()
```

This is a regional Manhattan plot.

The highest peaks represent candidate causal variants.

0.2.71 Step 10: Apply ABF Fine-Mapping

Define the ABF function used in the workshop.

```
run_abf <- function(  
  beta,  
  stderr,  
  W = 0.04  
) {  
  
  z <- beta/stderr  
  
  V <- stderr^2
```

```

r <- W/(W+V)

lbf <- (
  log(1-r) +
  r*(z^2)
) / 2

lbf_max <- max(lbf)

denom <- lbf_max +
  log(sum(
    exp(lbf-lbf_max)
  ))

prob <- exp(
  lbf-denom
)

return(prob)
}

```

Run ABF.

```

GWAS_df$PIP_ABF <-
  run_abf(
    GWAS_df$beta,
    GWAS_df$se
  )

```

0.2.72 Step 11: Visualize ABF PIPs

```

ggplot(
  GWAS_df,
  aes(
    SNP,
    PIP_ABF
  )
) +
  geom_point()

```

Interpretation:

Higher PIP
=

More likely causal

Often ABF will strongly prioritize only one signal.

0.2.73 Step 12: Run SuSiE

Compute LD matrix.

```
R <- cov(  
  scale(X)  
)
```

Run SuSiE.

```
susie_results <- susie_rss(  
  
  bhat = GWAS_df$beta,  
  
  shat = GWAS_df$se,  
  
  n = N,  
  
  R = R,  
  
  var_y = var(y),  
  
  L = 4,  
  
  estimate_residual_variance = TRUE  
)
```

This mirrors the workshop analysis.

0.2.74 Step 13: Extract PIPs

```
GWAS_df$PIP_SuSiE <-  
  susie_results$pip
```

View top variants.

```
head(  
  GWAS_df[,  
    order(  
      -GWAS_df$PIP_SuSiE
```

```
    ),
  ]
)
```

0.2.75 Step 14: Plot SuSiE PIPs

```
ggplot(
  GWAS_df,
  aes(
    SNP,
    PIP_SuSiE
  )
) +
geom_point()
```

You will often observe:

Several distinct peaks

corresponding to multiple causal variants.

0.2.76 Step 15: Examine Credible Sets

One of the most important outputs.

```
susie_results$sets$cs
```

Example:

\$L1

39

\$L2

20

\$L3

68

Interpretation:

Signal 1 → Variant 39

Signal 2 → Variant 20

Signal 3 → Variant 68

0.2.77 Comparing ABF and SuSiE

ABF may identify:

Variant 39

only.

SuSiE may identify:

Variant 20

Variant 39

Variant 68

because it allows multiple causal variants.

This is precisely why SuSiE has become the preferred modern approach.

0.3 Complete Workflow Summary

Simulate Genotypes



Simulate Phenotype



GWAS



ABF



PIP Estimates



SuSiE



Multiple Signals



Credible Sets



Candidate Causal Variants

0.3.1 Key Takeaways

A complete fine-mapping analysis begins with GWAS summary statistics and an LD matrix. ABF converts GWAS statistics into posterior probabilities under the assumption of a single causal variant. SuSiE extends this framework by modeling multiple independent signals within the same locus. The primary outputs are Posterior Inclusion Probabilities (PIPs) and Credible Sets, which identify the most likely causal variants. This workflow forms the foundation for downstream analyses such as eQTL mapping and colocalization.

0.3.2 Part 6: Understanding Credible Sets in Practice

After running a fine-mapping method such as SuSiE, researchers often focus immediately on the Posterior Inclusion Probabilities (PIPs).

However, the most important output is usually:

0.3.3 Credible Sets

Credible sets are the primary mechanism through which fine-mapping methods quantify uncertainty about causal variants.

Understanding credible sets properly is essential because most real GWAS loci do **not** resolve to a single variant.

The ISG workshop places substantial emphasis on interpreting credible sets and understanding why they may or may not contain the true causal variant.

0.3.4 Why Do We Need Credible Sets?

Suppose SuSiE reports:

```
rs39
PIP = 0.98
```

Life is easy.

We can confidently prioritize:

```
rs39
```

for downstream experiments.

Unfortunately, most loci look more like this:

rs39 PIP = 0.32

rs40 PIP = 0.28

rs41 PIP = 0.22

rs42 PIP = 0.10

rs43 PIP = 0.08

Now we face uncertainty.

Which SNP is truly causal?

Instead of pretending to know the answer, Bayesian fine-mapping explicitly represents this uncertainty using credible sets.

0.3.5 What Is a Credible Set?

A credible set is a collection of variants that jointly contain the causal variant with a specified probability.

Most commonly:

95% Credible Set

Meaning:

There is a 95% probability
that the causal SNP lies
within this set.

Notice the wording.

We are not saying:

95% probability
for every SNP.

We are saying:

95% probability
for the entire set.

0.3.6 Constructing a Credible Set

Suppose SuSiE produces:

rs39 PIP = 0.45

rs40 PIP = 0.25

rs41 PIP = 0.15

rs42 PIP = 0.08

rs43 PIP = 0.04

rs44 PIP = 0.03

Sort variants by decreasing PIP.

0.3.7 Step 1

Add the largest PIP.

rs39

Cumulative Probability = 0.45

Not enough.

0.3.8 Step 2

Add the second variant.

rs39 + rs40

0.45 + 0.25

= 0.70

Still not enough.

0.3.9 Step 3

Add the next SNP.

$$\begin{aligned} &rs39 + rs40 + rs41 \\ &= 0.85 \end{aligned}$$

Still below 95%.

0.3.10 Step 4

Add rs42.

$$0.93$$

Still below threshold.

0.3.11 Step 5

Add rs43.

$$0.97$$

Now we exceed:

$$95\%$$

Therefore:

$$\begin{aligned} &95\% \text{ Credible Set} \\ &= \\ &\{rs39, rs40, rs41, rs42, rs43\} \end{aligned}$$

0.3.12 Interpreting a Credible Set

This does NOT mean:

$$\text{All five SNPs are causal.}$$

Instead it means:

One (or more)
of these SNPs
is likely causal,
but the data cannot
distinguish them.

0.3.13 Ideal Credible Sets

The best possible outcome:

95% Credible Set

=
{rs39}

Single-SNP credible set.

This indicates:

Excellent resolution

and near-certain localization.

0.3.14 Moderate Credible Sets

Example:

95% Credible Set

=
{rs39, rs40, rs41}

This is still useful.

Experimental follow-up now requires testing:

3 variants

instead of:

500 variants

0.3.15 Poor Credible Sets

Sometimes:

95% Credible Set

=

50 SNPs

or even:

100 SNPs

This indicates:

Poor fine-mapping resolution

The data cannot isolate the causal variant.

0.3.16 Why Do Large Credible Sets Occur?

Several factors contribute.

0.3.17 Reason 1: High Linkage Disequilibrium

Suppose:

rs39

rs40

rs41

rs42

all have:

$r^2 > 0.95$

The data cannot distinguish them.

Result:

Large Credible Set

0.3.18 Reason 2: Small Sample Size

Imagine:

$$N = 500$$

instead of:

$$N = 50000$$

Less information is available.

PIPs become more diffuse.

Credible sets become larger.

The workshop explicitly demonstrates how increasing sample size improves fine-mapping resolution.

0.3.19 Reason 3: Weak Genetic Effects

Suppose:

$$\beta = 0.02$$

instead of:

$$\beta = 0.20$$

The signal becomes harder to localize.

Again:

Larger Credible Sets

0.3.20 Example: Excellent Fine-Mapping

rs39 0.97

rs40 0.01

rs41 0.01

rs42 0.01

Credible Set:

{rs39}

Interpretation:

Near-perfect localization

0.3.21 Example: Difficult Fine-Mapping

rs39 0.25

rs40 0.24

rs41 0.22

rs42 0.16

rs43 0.13

Credible Set:

{rs39, rs40, rs41, rs42, rs43}

Interpretation:

Strong uncertainty

0.3.22 Multiple Credible Sets

Modern loci often contain multiple signals.

Example:

Signal 1

CS1 = {rs20}

and

Signal 2

CS2 = {rs39, rs40}

and

Signal 3

CS3 = {rs68}

This is exactly what SuSiE was designed to discover.

Each credible set corresponds to a separate causal signal.

0.3.23 Viewing Credible Sets in SuSiE

After fitting SuSiE:

```
susie_results$sets$cs
```

Example output:

\$L1
39

\$L2
20

\$L3
68

Interpretation:

Three independent signals

located at:

Variant 39

Variant 20

Variant 68

The workshop uses this exact output structure.

0.3.24 Credible Set Coverage

Most studies report:

95% coverage

but other thresholds are possible.

Examples:

80%

90%

95%

99%

0.3.25 Trade-Off

Higher coverage:

99%

gives:

More certainty

but:

Larger Credible Sets

Lower coverage:

80%

gives:

Smaller Credible Sets

but:

Greater risk
of missing the causal SNP

0.3.26 The Workshop Example

The workshop explores changing credible set coverage from:

95%

to

80%

to observe how credible sets shrink as the confidence threshold decreases.

0.3.27 Credible Sets Versus Lead SNPs

Researchers often report:

Lead SNP

which is simply:

Smallest p-value

or

Highest PIP

However:

Lead SNP

≠

Credible Set

The credible set captures uncertainty.

The lead SNP does not.

This distinction is extremely important.

0.3.28 What Should Be Reported in a Paper?

A modern fine-mapping study should report:

1. Lead SNP
2. PIP
3. Credible Set
4. Credible Set Size
5. Fine-mapping Method
6. LD Reference Panel

Example:

Lead Variant:

rs39

PIP:

0.94

95% Credible Set:

{rs39, rs40}

Credible Set Size:

2

This provides much more information than a p-value alone.

0.3.29 Biological Interpretation

Suppose:

Credible Set Size = 1

Excellent candidate for:

- CRISPR validation
 - Functional assays
 - Reporter assays
-

Suppose:

Credible Set Size = 50

Additional data may be required:

- Larger GWAS
- Better LD panel
- eQTL analysis
- Colocalization
- Functional annotations

to refine the signal.

0.3.30 Key Takeaways

Credible sets are one of the most important outputs of modern fine-mapping methods. Rather than identifying a single causal variant, they quantify uncertainty by reporting a set of variants that collectively contain the causal SNP with a specified probability, typically 95%. Small credible sets indicate strong fine-mapping resolution, while large credible sets reflect uncertainty caused by high LD, limited sample size, or weak genetic effects. In modern statistical genetics, credible sets are often more informative than lead SNPs because they explicitly represent the uncertainty inherent in causal variant identification.

0.3.31 Part 7: The Impact of Sample Size, Effect Size, and Linkage Disequilibrium on Fine-Mapping Resolution

One of the most important lessons in fine-mapping is:

The quality of your fine-mapping results depends heavily on your data.

Researchers often ask:

Why is my credible set so large?

Why are my PIPs so low?

Why can't I identify the causal SNP?

The answer usually involves three factors:

1. Sample Size
2. Linkage Disequilibrium (LD)
3. Effect Size

The ISG workshop specifically explores these factors through simulation experiments.

0.3.32 Fine-Mapping Resolution

Resolution refers to how precisely we can localize a causal variant.

Excellent resolution:

95% Credible Set

=

1 SNP

Poor resolution:

95% Credible Set

=

50 SNPs

Our goal is always:

Small Credible Sets

High PIPs

0.3.33 Factor 1: Sample Size

The single most important factor in fine-mapping is:

Sample Size

Suppose we conduct a GWAS with:

$$N = 500$$

individuals.

The statistical signal will be noisy.

Now imagine:

$$N = 500,000$$

The estimates become much more precise.

0.3.34 Why Sample Size Matters

The standard error of a regression coefficient is approximately:

$$SE \propto 1/\sqrt{N}$$

Meaning:

Larger N
↓
Smaller SE
↓
Larger Z-scores
↓
Higher PIPs

0.3.35 Example

Suppose the true effect size is:

$$\beta = 0.10$$

With:

$$N = 500$$

we might observe:

$$\beta = 0.10$$

$$SE = 0.08$$

$$Z = 1.25$$

Weak evidence.

With:

$$N = 50,000$$

we might observe:

$$\beta = 0.10$$

$$SE = 0.01$$

$$Z = 10$$

Very strong evidence.

0.3.36 Simulation Example

The workshop increases the sample size from:

$$N = 1000$$

to

$$N = 5000$$

while keeping the same causal variants.

The result:

Higher PIPs
Better localization
Smaller credible sets

0.3.37 Visual Intuition

Small study:

```

      *
    * *
  * * *
* * * *
```

Noisy signal.

Large study:

```

  *
  *
  *
  *
```

Sharper signal.

0.3.38 Practical Rule

If you double the sample size:

Information increases

If you increase sample size by:

10×

fine-mapping performance often improves dramatically.

0.3.38.1 Factor 2: Linkage Disequilibrium (LD)

The second major factor is:

LD

Fine-mapping is easiest when SNPs are weakly correlated.

0.3.39 Low LD Example

rs39 ---- rs40 ---- rs41

Correlations:

$$r^2 = 0.05$$

Each SNP behaves independently.

Fine-mapping can easily identify:

rs39

if it is causal.

0.3.40 High LD Example

rs39 ---- rs40 ---- rs41

Correlations:

$$r^2 = 0.99$$

Now:

rs39
rs40
rs41

look almost identical statistically.

0.3.41 Consequence

The model cannot distinguish:

Which SNP is causal?

Result:

Large Credible Sets

0.3.42 Workshop Demonstration

The workshop increases LD from:

0.3

to

0.95

while keeping sample size fixed.

Result:

PIPs decrease

Credible sets grow

Resolution worsens

0.3.43 Why LD Is So Difficult

Suppose:

rs39

is causal.

If:

$$r^2(rs39, rs40) = 0.99$$

then:

rs40

will have almost the same GWAS statistics.

Example:

$$rs39 \quad p = 1 \times 10^{-12}$$

$$rs40 \quad p = 2 \times 10^{-12}$$

Both look causal.

The data cannot distinguish them.

0.3.44 Fine-Mapping Resolution Spectrum

Low LD:

Credible Set

=

{rs39}

Excellent.

Moderate LD:

Credible Set

=

{rs39, rs40, rs41}

Acceptable.

Very High LD:

Credible Set

=

25 SNPs

Poor.

0.3.45 Factor 3: Effect Size

The third important factor is:

Effect Size

Some variants have strong biological effects.

Others have very small effects.

0.3.46 Large Effect Variant

$$\beta = 0.25$$

Produces:

Large Z-score

High PIP

Often:

Single-SNP Credible Set

0.3.47 Small Effect Variant

$$\beta = 0.01$$

Produces:

Small Z-score

Diffuse PIPs

Result:

Large Credible Set

0.3.48 Genetic Architecture Matters

Traits differ dramatically.

Examples:

LDL Cholesterol

often contains:

Large effects

Examples:

Educational Attainment

often contains:

Very small effects

Consequently:

Some traits are naturally
easier to fine-map.

0.3.49 Heritability and Fine-Mapping

The workshop simulates traits with:

$$h^2 = 0.10$$

and suggests reducing:

$$h^2 = 0.01$$

to observe what happens.

When heritability decreases:

Signals weaken

PIPs decrease

Credible sets increase

0.3.50 Interaction Between Factors

These factors do not operate independently.

0.3.51 Best Case Scenario

Large Sample Size

Low LD

Large Effects

Result:

Tiny Credible Sets

High PIPs

0.3.52 Worst Case Scenario

Small Sample Size

High LD

Tiny Effects

Result:

Huge Credible Sets

Low PIPs

0.3.53 Real-World Example

Suppose a locus contains:

50 SNPs

in strong LD.

Even with:

$N = 100,000$

fine-mapping may still struggle.

Why?

Because LD creates a fundamental statistical limitation.

More data helps, but cannot completely eliminate the problem.

0.3.54 Why Different Ancestries Matter

Different populations have different LD structures.

European ancestry:

Long LD blocks

Some regions difficult to resolve.

African ancestry:

Shorter LD blocks

Often provides better fine-mapping resolution.

This is one reason multi-ancestry fine-mapping has become increasingly popular.

0.3.55 Simulation Function from the Workshop

The workshop creates a function:

```
simulate_and_finemap(  
  N_sim,  
  LD_baseline  
)
```

allowing researchers to change:

Sample Size

LD

Heritability

and directly observe how PIPs and credible sets change.

This is an excellent exercise for understanding fine-mapping behavior.

0.3.56 Practical Lessons

When fine-mapping performs poorly, ask:

0.3.57 Is the sample size large enough?

More samples

=

Better resolution

0.3.58 Is LD too high?

High LD

=

Harder localization

0.3.59 Are effect sizes too small?

Small effects

=

Lower PIPs

0.3.60 Is the LD reference panel appropriate?

Wrong ancestry

=

Potentially misleading results

0.3.61 Common Beginner Mistake

Many people assume:

Fine-mapping failed.

when they see:

Credible Set Size = 50

In reality:

The method may be working perfectly.

The data simply may not contain enough information to distinguish the variants.

This is an important mindset shift.

Fine-mapping quantifies uncertainty.

It does not magically eliminate uncertainty.

0.3.62 Key Takeaways

Fine-mapping resolution is determined primarily by sample size, linkage disequilibrium, and effect size. Larger sample sizes produce more precise effect estimates and higher PIPs. High LD makes neighboring variants statistically indistinguishable and often leads to larger credible sets. Strong genetic effects are easier to localize than weak effects. Even the best fine-mapping method cannot overcome severe LD or limited information. Understanding these factors is essential for interpreting fine-mapping results realistically and avoiding overconfidence in causal variant claims.

0.3.63 Part 8: Fine-Mapping with Individual-Level Data vs Summary Statistics

One of the first questions in any fine-mapping project is:

What data do I actually have?

The answer determines:

- Which fine-mapping method you can use
- How accurate your results may be
- How LD should be estimated
- Whether colocalization is possible
- Which branch of the fine-mapping workflow you should follow

This is exactly the starting point of the ISG fine-mapping decision tree.

0.3.64 Why Data Availability Matters

In an ideal world, every researcher would have:

Genotypes
Phenotypes
Covariates

for every participant.

Unfortunately, this is rarely the case.

Most published GWAS provide only:

Summary Statistics

such as:

SNP
Beta
SE
P-value
Alleles

As a result, modern fine-mapping methods must work with multiple types of data.

0.3.65 Two Main Types of Data

0.3.66 Individual-Level Data

You have access to:

Genotypes
Phenotypes
Covariates

for each person.

Example:

Person	SNP1	SNP2	SNP3	Phenotype
1	0	1	2	3.2
2	1	1	0	1.7
3	2	0	1	4.5

0.3.67 Summary Statistics

You only have GWAS results.

Example:

SNP	Beta	SE	P-value
rs1	0.12	0.03	1e-8
rs2	0.10	0.03	5e-7
rs3	0.02	0.04	0.40

0.4 No individual genotypes are available.

0.4.1 Why Summary Statistics Became Popular

Large GWAS datasets are often restricted.

Examples:

- UK Biobank
- FinnGen

- deCODE
- iPSYCH

Sharing raw genotypes creates:

- Privacy concerns
- Storage challenges
- Regulatory issues

Therefore most studies release:

Summary Statistics

instead.

0.4.2 Individual-Level Data Workflow

Suppose you have:

Genotypes
Phenotypes

The workflow becomes:

```
Raw Genotypes
  ↓
Quality Control
  ↓
GWAS
  ↓
Compute LD Directly
  ↓
Fine-Mapping
```

This is the gold standard.

0.4.3 Why Individual-Level Data Is Powerful

Because you can calculate:

LD

directly from the study participants.

Example:

```
R <- cov(  
  scale(X)  
)
```

where:

X

is the genotype matrix.

This produces:

In-sample LD

which is the most accurate LD possible.

0.4.4 What Is In-Sample LD?

Suppose your GWAS contains:

10,000 Finnish individuals

You calculate LD using exactly those individuals.

Result:

Perfectly matched LD

This is called:

In-Sample LD

and is considered the gold standard.

0.4.5 Summary Statistics Workflow

Now suppose you only have:

Beta
SE
P-values

The workflow becomes:

Summary Statistics
↓
External LD Panel
↓
Fine-Mapping

Now LD must come from somewhere else.

0.4.6 External LD Reference Panels

Common reference panels include:

- 1000 Genomes
- UK Biobank
- TOPMed
- HRC

These provide genotype information for estimating LD.

0.4.7 Example

Suppose your GWAS contains:

European ancestry individuals

You may estimate LD from:

1000 Genomes Europeans

instead.

0.4.8 Why LD Matching Matters

This is one of the most important concepts in fine-mapping.

Suppose:

GWAS

contains:

Finnish ancestry

but LD comes from:

African ancestry

Then:

LD structure differs

which can distort fine-mapping results.

0.4.9 Example

True GWAS LD:

rs39 ↔ rs40

$r^2 = 0.95$

Reference LD:

rs39 ↔ rs40

$r^2 = 0.30$

These tell completely different stories.

The fine-mapping method may become confused.

0.4.10 Consequences of LD Mismatch

Poorly matched LD can cause:

False signals

Incorrect PIPs

Wrong credible sets

Failed colocalization

This is one of the most common causes of unreliable fine-mapping.

0.4.11 Why SuSiE Requires Good LD

Recall that SuSiE models:

Multiple causal variants

To separate these signals, it must know:

Which SNPs are correlated?

Therefore:

Good LD

=

Good Fine-Mapping

The workshop repeatedly emphasizes using accurate in-sample LD whenever possible.

0.4.12 Methods That Need LD

Examples:

SuSiE

FINEMAP

CAVIAR

CARMA

All require LD information.

Without LD they cannot properly model correlation among SNPs.

0.4.13 Methods That Can Work Without LD

Some methods use only summary statistics.

Example:

ABF

The workshop ABF example requires:

Beta

SE

but not an LD matrix.

This is one reason ABF remains useful.

0.4.14 Individual-Level Data Branch

The workshop flowchart recommends:

Individual-Level Data



Single Cohort



Compute In-Sample LD



SuSiE

This is typically the preferred route.

0.4.15 Summary Statistics Branch

If only GWAS summary statistics are available:

Summary Statistics



Reference LD



SuSiE

FINEMAP

CARMA

depending on the study design.

0.4.16 Single-Cohort GWAS

Example:

FinnGen

or

UK Biobank

analyzed independently.

Workflow:

Single Cohort

↓

Ancestry-Matched LD

↓

Fine-Mapping

This is generally straightforward.

0.4.17 Meta-Analysis GWAS

Now consider:

European Cohort

+

Finnish Cohort

+

Japanese Cohort

+

African Cohort

combined into one GWAS.

This creates a challenge.

0.4.18 Why Meta-Analysis Is Difficult

Each cohort has:

Different LD

Therefore:

One LD Matrix

may no longer represent the data accurately.

This is one reason specialized meta-analysis fine-mapping methods were developed.

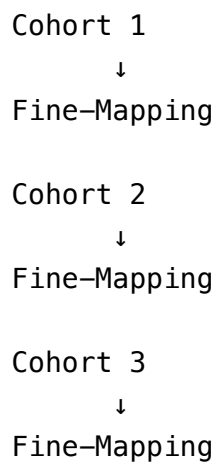
0.4.19 Cohort-Specific Fine-Mapping

A common strategy is:

Fine-map each cohort separately

then compare results.

Workflow:



Then integrate results.

0.4.20 Multi-Ancestry Fine-Mapping

An increasingly popular approach is:

Multi-Ancestry Fine-Mapping

because different ancestries have different LD structures.

Example:

European LD:

rs39 rs40 rs41 rs42

all highly correlated.

African LD:

rs39

more clearly separated.

Combining information can dramatically improve resolution.

0.4.21 Practical Example

Imagine:

European GWAS

produces:

Credible Set Size = 40

African GWAS:

Credible Set Size = 5

The African LD structure may help isolate the causal variant.

0.4.22 Which Data Type Is Best?

Ranking:

0.4.23 Best

Individual-Level Data

+

In-Sample LD

0.4.24 Good

Summary Statistics

+

Matched LD Panel

0.4.25 Risky

Summary Statistics

+

Poorly Matched LD

0.4.26 Worst

Summary Statistics

+

No LD Information

0.4.27 Practical Advice

Whenever possible:

0.4.28 Use in-sample LD

Best choice

0.4.29 Match ancestry

European GWAS

→ European LD

African GWAS

→ African LD

0.4.30 Avoid mixing populations

unless using methods designed for multi-ancestry analyses.

0.4.31 Connection to Colocalization

Everything we learn here becomes even more important later.

Methods such as:

`coloc.susie`

require accurate fine-mapping results.

Poor LD estimation can lead to:

False colocalization

or

Missed colocalization

Therefore:

Good LD

=

Good Fine-Mapping

=

Good Colocalization

0.4.32 Key Takeaways

The type of data available determines the entire fine-mapping strategy. Individual-level data provide the highest-quality analyses because LD can be computed directly from the study participants. When only summary statistics are available, an external LD reference panel must be used. Accurate ancestry matching between the GWAS cohort and the LD reference panel is critical because LD mismatch can distort PIPs, credible sets, and downstream analyses. Modern methods such as SuSiE, FINEMAP, and CARMA depend heavily on accurate LD information, making LD quality one of the most important determinants of successful fine-mapping.

0.4.33 Part 9: Major Fine-Mapping Methods — ABF, SuSiE, FINEMAP, CAVIAR, CARMA, and FastMap

By now, you understand the core concepts behind fine-mapping:

- Linkage Disequilibrium (LD)
- Posterior Inclusion Probabilities (PIPs)
- Credible Sets
- Bayesian Inference
- Multiple Causal Variants

The next question naturally becomes:

Which fine-mapping method should I use?

Modern statistical genetics offers several methods, each designed for different situations.

The ISG workshop flowchart includes methods such as:

ABF
SuSiE
FINEMAP
CARMA
FastMap

Choosing the correct method depends largely on:

Available Data
LD Information
Study Design
Computational Resources

0.4.34 The Evolution of Fine-Mapping Methods

Historically:

GWAS
↓
ABF
↓
CAVIAR
↓

FINEMAP

↓

SuSiE

↓

CARMA

↓

FastMap

Each generation addressed limitations of earlier methods.

0.4.35 1. Approximate Bayes Factor (ABF)

0.4.36 Main Idea

Assume:

One causal variant
per locus

Compute:

Bayes Factors

for each SNP.

Convert them into:

Posterior Probabilities

0.4.37 Inputs

Beta
SE

Only GWAS summary statistics are required.

No LD matrix is needed.

0.4.38 Advantages

Fast
Simple

Minimal inputs

0.4.39 Disadvantages

Single causal variant assumption

Cannot properly model:

Multiple signals

0.4.40 Best Use Case

When:

No trustworthy LD matrix exists

or

Quick exploratory analysis

0.4.41 Example

Methods based on ABF:

Wakefield ABF

coloc.abf

0.4.42 2. CAVIAR

One of the earliest methods designed to handle:

Multiple causal variants

0.4.43 Main Idea

Instead of assuming:

1 causal SNP

CAVIAR models:

Several causal SNPs

simultaneously.

0.4.44 Inputs

Z scores

LD matrix

0.4.45 Outputs

Causal probabilities

Credible sets

0.4.46 Advantages

Accounts for LD.

Allows multiple causal variants.

0.4.47 Disadvantages

Can become computationally expensive.

Performance decreases in very large regions.

0.4.48 Historical Importance

CAVIAR was one of the first major advances beyond ABF.

Many later methods were inspired by it.

0.4.49 3. FINEMAP

FINEMAP became popular because it is:

Fast

Accurate

Scalable

for large GWAS datasets.

0.4.50 Main Idea

Uses Bayesian model selection.

Searches through possible causal configurations.

Example:

Model 1:
rs39

Model 2:
rs20 + rs39

Model 3:
rs20 + rs39 + rs68

and evaluates which model best explains the data.

0.4.51 Inputs

Summary Statistics
LD Matrix

0.4.52 Outputs

PIPs

Credible Sets

Causal Configurations

0.4.53 Advantages

Handles:

Multiple causal variants

efficiently.

Widely used.

Very accurate.

0.4.54 Disadvantages

Requires:

Good LD

Performance may degrade if LD is poorly estimated.

0.4.55 Best Use Cases

Large GWAS.

High-resolution fine-mapping.

Biobank-scale studies.

0.4.56 4. SuSiE (Sum of Single Effects)

SuSiE has become one of the most widely used methods in modern statistical genetics.

0.4.57 Main Idea

Model phenotype as:

Signal 1
+
Signal 2
+
Signal 3

rather than:

One signal

0.4.58 Inputs

Summary Statistics

LD Matrix

or

Individual-level data

0.4.59 Outputs

PIPs

Credible Sets

for each independent signal.

0.4.60 Advantages

Excellent handling of:

Multiple signals

Produces interpretable credible sets.

Computationally efficient.

Works well with colocalization.

0.4.61 Disadvantages

Requires:

Accurate LD

Sensitive to LD mismatch.

0.4.62 Why SuSiE Is Popular

SuSiE became popular because it combines:

Accuracy
Interpretability
Speed

while naturally handling multiple causal variants.

0.4.63 5. CARMA

CARMA stands for:

Causal Variants Identification
in the Presence of Allelic Heterogeneity
and Outliers

CARMA is a relatively recent method.

0.4.64 Motivation

Traditional methods often assume:

Summary statistics
contain no major errors.

Real GWAS may contain:

Outlier SNPs

Genotyping artifacts

Imputation errors

These can distort fine-mapping.

0.4.65 Main Idea

CARMA explicitly models:

Outliers

and attempts to remain robust when data quality is imperfect.

0.4.66 Inputs

Summary Statistics

LD Matrix

0.4.67 Advantages

Robust to:

Outlier SNPs

Works well in challenging datasets.

Often improves credible set quality.

0.4.68 Disadvantages

More computationally intensive.

More complex to interpret.

0.4.69 Best Use Cases

When:

Data quality is uncertain

or

Large meta-analysis studies

where artifacts may occur.

0.4.70 6. FastMap

FastMap is one of the newest methods appearing in modern fine-mapping workflows.

0.4.71 Motivation

Biobank-scale datasets now contain:

Millions of individuals

and

Thousands of loci

Traditional methods become computationally expensive.

0.4.72 Main Idea

Use efficient approximations to dramatically speed up fine-mapping.

0.4.73 Advantages

Very fast
Scalable

Suitable for:

Thousands of loci

simultaneously.

0.4.74 Disadvantages

Still relatively new.

Less extensively validated than SuSiE or FINEMAP.

0.4.75 Best Use Cases

Large-scale biobank projects.

Meta-analysis fine-mapping.

Population-scale studies.

0.4.76 Comparison Table

Method	Multiple Signals	Needs LD	Speed	Modern Usage
ABF	No	No	Very Fast	Moderate

Method	Multiple Signals	Needs LD	Speed	Modern Usage
CAVIAR	Yes	Yes	Moderate	Historical
FINEMAP	Yes	Yes	Fast	High
SuSiE	Yes	Yes	Fast	Very High
CARMA	Yes	Yes	Moderate	Growing
FastMap	Yes	Yes	Very Fast	Emerging

0.4.77 Which Method Should You Use?

0.4.77.1 Scenario 1

You only have:

Beta
SE

No LD matrix.

Use:

ABF

0.4.78 Scenario 2

You have:

Summary Statistics
+
Good LD

Use:

SuSiE

or

FINEMAP

0.4.79 Scenario 3

You suspect:

Outliers

Use:

CARMA

0.4.80 Scenario 4

You need:

Very large-scale analysis

Use:

FastMap

0.4.81 Scenario 5

You plan to perform:

Colocalization

Use:

SuSiE

because it integrates naturally with:

coloc.susie

0.4.82 The Current Consensus

If you ask statistical geneticists today:

What is the default fine-mapping method?

the most common answer is:

SuSiE

because it provides:

- Multiple signal modeling
- Credible sets
- PIPs
- Strong theoretical foundations
- Excellent software support

However:

FINEMAP

remains extremely popular and is often used alongside SuSiE.

0.4.83 A Practical Recommendation

For most researchers:

0.4.83.1 Individual-Level Data

SuSiE

0.4.83.2 Summary Statistics + Good LD

SuSiE or FINEMAP

0.4.84 No LD Available

ABF

0.4.84.1 Suspected Data Artifacts

CARMA

0.4.84.2 Massive Biobank Analyses

FastMap

0.4.85 Fine-Mapping Methods in the Bigger Picture

All these methods aim to answer the same question:

Which variants are most likely causal?

Their differences lie mainly in:

- Assumptions
- Computational strategy
- Treatment of multiple signals
- Robustness to imperfect data

The core outputs remain:

Posterior Inclusion Probabilities

Credible Sets

which are the language of modern fine-mapping.

0.4.86 Key Takeaways

Several fine-mapping methods are available, each designed for different scenarios. ABF is simple and useful when LD information is unavailable but assumes a single causal variant. CAVIAR was one of the first methods to model multiple causal variants. FINEMAP and SuSiE are currently among the most widely used methods because they efficiently model multiple signals and produce credible sets. CARMA extends fine-mapping by explicitly handling outlier variants, while FastMap focuses on computational scalability for biobank-sized datasets. In

practice, SuSiE and FINEMAP have become the dominant choices for modern fine-mapping studies when reliable LD information is available.

0.4.87 Part 10: The Fine-Mapping Decision Tree — Choosing the Correct Analysis Pipeline

One of the biggest challenges for beginners is not running fine-mapping software.

The real challenge is:

Choosing the correct workflow.

Many researchers immediately ask:

Should I use SuSiE?

Should I use FINEMAP?

Do I need CARMA?

Can I run coloc?

What LD matrix should I use?

The answer depends entirely on:

What data are available.

This is why the ISG workshop introduced a Fine-Mapping Decision Tree.

Rather than choosing a method first, we should first determine:

What information do we have?

0.4.88 The Big Picture

Every fine-mapping project starts with:

Available Data

↓

Appropriate LD

↓

Fine-Mapping Method



Credible Sets



Biological Interpretation

The workflow branches depending on the data type.

0.4.89 Step 1: Do You Have Individual-Level Data?

The first question is:

Do you have access to genotypes and phenotypes?

Examples:

PLINK files

.bed
.bim
.fam

or

VCF files

Phenotype files

If yes:

Go down the
Individual-Level Data branch

If no:

Go down the
Summary Statistics branch

0.4.90 Branch A: Individual-Level Data

This is the ideal situation.

You have:

Genotypes

Phenotypes

Covariates

Example datasets:

UK Biobank

FinnGen

iPSYCH

All of Us

when individual-level access is granted.

0.4.91 Why Individual-Level Data Is Valuable

Because you can calculate:

In-Sample LD

directly from the study participants.

Workflow:

Individual-Level Data



Quality Control



GWAS



Compute In-Sample LD



SuSiE



Credible Sets

This is usually the preferred strategy.

0.4.92 Single-Cohort Individual-Level Data

Suppose all samples come from:

One Cohort

Example:

FinnGen

or

UK Biobank

alone.

Workflow:

Single Cohort

↓

In-Sample LD

↓

SuSiE

This generally provides the most reliable fine-mapping.

0.4.93 Multi-Cohort Individual-Level Data

Suppose you have:

European Cohort

African Cohort

Asian Cohort

with full genotypes.

Now you have two options.

0.4.94 Option 1: Fine-Map Separately

European



Fine-Map

African



Fine-Map

Asian



Fine-Map

Then compare results.

This is often recommended.

0.4.95 Option 2: Multi-Ancestry Fine-Mapping

Combine all cohorts.

Use specialized approaches.

Advantages:

Different LD structures
improve resolution

This has become increasingly popular in modern statistical genetics.

0.4.96 Branch B: Summary Statistics Available

Most researchers fall into this category.

You have:

Beta

SE

P-value

but not raw genotypes.

Workflow:

Summary Statistics



LD Reference Panel



Fine-Mapping

0.4.97 The Critical Question

The next question becomes:

Do you have a good LD reference panel?

This is one of the most important decisions in fine-mapping.

0.4.98 Scenario B1: Good LD Available

Example:

European GWAS

European LD Panel

or

Finnish GWAS

Finnish LD Panel

Excellent.

Workflow:

Summary Statistics



Matched LD



SuSiE

or

FINEMAP

This is the most common workflow today.

0.4.99 Why Matching Matters

Suppose:

GWAS = European

but

LD = African

Now:

LD structure differs

which may distort:

PIPs

Credible Sets

and potentially produce misleading results.

0.4.100 Scenario B2: No Reliable LD Available

Sometimes you only have:

Beta

SE

and no trustworthy LD.

In this situation:

SuSiE

is not appropriate because it requires LD.

Instead:

ABF

becomes useful.

Workflow:

Summary Statistics



ABF



Posterior Probabilities

Remember:

ABF assumes
one causal variant
per locus

so this is a compromise.

0.4.101 Scenario B3: Suspected Data Problems

Sometimes summary statistics contain:

Imputation errors

Outliers

Meta-analysis artifacts

These can distort fine-mapping.

Workflow:

Summary Statistics



LD Panel



CARMA

CARMA was specifically developed to improve robustness to problematic variants.

0.4.102 Meta-Analysis GWAS Branch

This is where things become complicated.

Suppose your GWAS combines:

UK Biobank

FinnGen

deCODE

Biobank Japan

into a single meta-analysis.

Now:

Multiple LD structures

exist simultaneously.

0.4.103 Why Meta-Analysis Is Challenging

Fine-mapping methods assume:

One LD matrix

but meta-analysis combines:

Many LD matrices

This creates a mismatch.

0.4.104 Strategy 1: Cohort-Specific Fine-Mapping

If cohort-specific summary statistics are available:

FinnGen

↓

Fine-Map

UK Biobank



Fine-Map

deCODE



Fine-Map

Then integrate results.

This is often the preferred strategy.

0.4.105 Strategy 2: Specialized Meta-Analysis Methods

If only meta-analysis results are available:

Use methods such as:

FastMap

or related approaches designed for meta-analysis settings.

These methods attempt to account for heterogeneous LD structures.

0.4.106 The Colocalization Branch

Eventually many projects move beyond fine-mapping.

Suppose you identify:

GWAS Signal

and

eQTL Signal

in the same region.

The next question becomes:

Are they driven by
the same causal variant?

This is where:

Colocalization

begins.

If you used ABF, you can use,

`coloc.abf`

Workflow:

```
GWAS Summary Statistics
      +
eQTL Summary Statistics
      ↓
coloc.abf
```

Assumption:

One causal variant
per locus

If You used SuSiE You can use:

`coloc.susie`

Workflow:

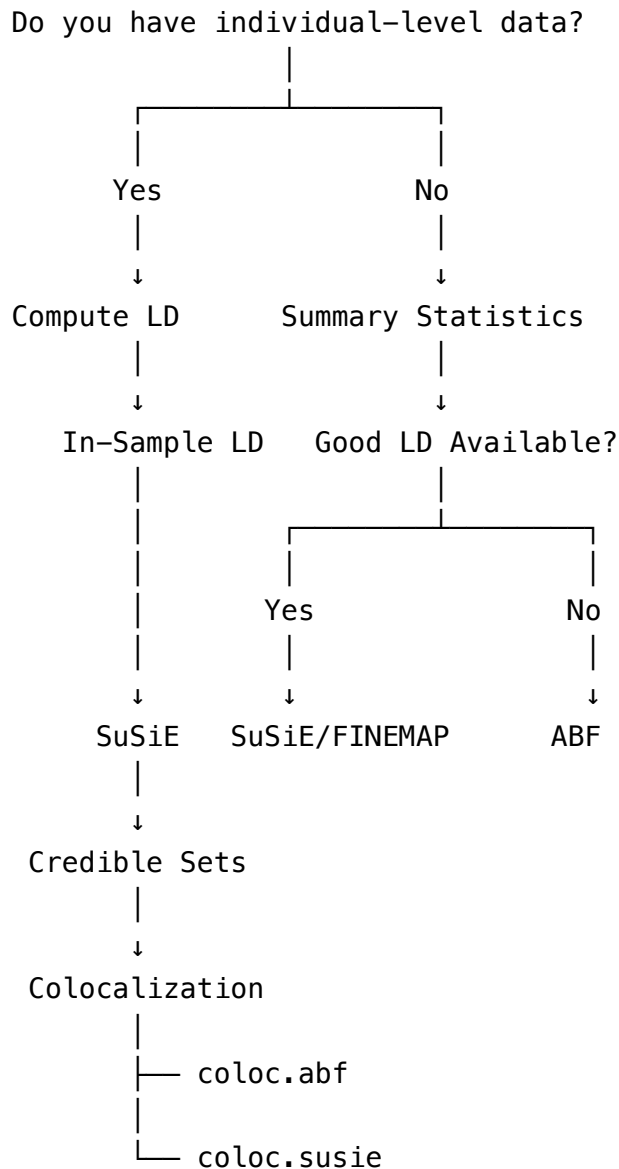
```
GWAS Fine-Mapping
      +
eQTL Fine-Mapping
      ↓
coloc.susie
```

This is generally preferred because it allows:

Multiple causal variants

and aligns naturally with modern fine-mapping.

0.4.107 The Complete Decision Tree



0.4.108 Example 1: FinnGen Study

You have:

FinnGen Summary Statistics

and

FinnGen LD Panel

Recommendation:

SuSiE

0.4.109 Example 2: Public GWAS Catalog Result

You have:

Beta

SE

but no LD information.

Recommendation:

ABF

0.4.110 Example 3: UK Biobank Genotypes

You have:

Raw Genotypes

Recommendation:

Compute In-Sample LD

Run SuSiE

0.4.111 Example 4: Psychiatric Genomics Consortium Meta-Analysis

You have:

Meta-analysis summary statistics

Recommendation:

Cohort-specific fine-mapping

if possible.

Otherwise:

FastMap

or another meta-analysis-aware approach.

0.4.112 Common Beginner Mistake

Many people start with:

Which method should I use?

The better question is:

What data do I have?

The data determine the method.

Not the other way around.

0.4.113 Recommended Default Workflow

For most modern studies:

```
Summary Statistics
      +
Matched LD
      ↓
SuSiE
      ↓
Credible Sets
      ↓
eQTL Integration
      ↓
coloc.susie
```

This has become one of the most common pipelines in statistical genetics.

0.4.114 Key Takeaways

The choice of fine-mapping method should be driven by the available data rather than personal preference. Individual-level data with in-sample LD provide the

highest-quality analyses and are ideally suited for SuSiE. When only summary statistics are available, matched LD reference panels become critical. SuSiE and FINEMAP are generally preferred when reliable LD is available, while ABF remains useful when LD information is unavailable. Meta-analysis studies introduce additional complexity because LD differs across cohorts, often requiring cohort-specific fine-mapping or specialized methods such as FastMap. Ultimately, fine-mapping serves as the foundation for downstream analyses such as eQTL integration and colocalization.

0.4.115 Part 11: Integrating Fine-Mapping with eQTL Data — Identifying the Target Gene Behind a GWAS Signal

At this point, we have learned how to:

```
Run GWAS
  ↓
Fine-Map Loci
  ↓
Identify Candidate Causal Variants
```

However, a major biological question remains:

`text id="mns4g9"` Which gene is affected by the causal variant?

Finding a causal SNP is only part of the story.

Ultimately, most researchers want to know:

`text id="76b7s3"` Which gene drives disease risk?

This is where eQTL analysis becomes one of the most powerful tools in statistical genetics.

0.4.116 The Main Problem in GWAS

Most GWAS hits are:

`text id="65c65s"` Non-coding variants

rather than protein-coding mutations.

For example:

`text id="rpkfwy"` rs12345

may lie:

```text id="js9bpa" 50 kb upstream

or

100 kb downstream

of a gene.

The SNP itself does not tell us:

```text id="2m7cti"

Which gene is affected.

0.4.117 What Is an eQTL?

An eQTL is a:

0.4.117.1 Expression Quantitative Trait Locus

A genetic variant that influences gene expression.

Example:

rs12345

may increase expression of:

Gene A

while decreasing expression of:

Gene B

0.4.118 Conceptually

Genotype

↓

Gene Expression

↓
Disease Risk

This creates a biological mechanism linking DNA variation to disease.

0.4.119 Why eQTLs Are Important

Suppose a schizophrenia GWAS identifies:

rs12345

Fine-mapping shows:

PIP = 0.95

Excellent.

But we still do not know:

Which gene
is responsible.

Now suppose:

rs12345

is also an eQTL for:

CACNA1C

This provides a biological hypothesis:

Variant
↓
CACNA1C Expression
↓
Disease Risk

0.4.120 Where Do eQTL Data Come From?

Several large projects have measured gene expression and genotype data simultaneously.

Popular resources include:

- GTEx
- eQTL Catalogue
- PsychENCODE
- CommonMind Consortium
- eQTLGen

These datasets allow researchers to ask:

Does this SNP affect
gene expression?

0.4.121 Cis-eQTLs

Most studies focus on:

0.4.121.1 Cis-eQTLs

Variants located near the gene they regulate.

Typically:

Within 1 Mb

of the gene.

Example:

rs12345

affects:

Gene A

located:

200 kb away

0.4.122 Trans-eQTLs

Some variants regulate distant genes.

Example:

`text id="z1f92d" Chromosome 1 SNP`

affecting:

`text id="lmvm4n" Chromosome 12 Gene`

These are called:

0.4.122.1 Trans-eQTLs

and are generally harder to detect.

0.4.123 The Fine-Mapping + eQTL Workflow

Modern statistical genetics often follows:

`text id="jlmq29" GWAS ↓ Fine-Mapping ↓ Candidate SNPs ↓ eQTL
Database ↓ Candidate Genes`

0.4.124 Example

Suppose fine-mapping produces:

`text id="d6mqt4" 95% Credible Set`

`= rs39 rs40 rs41`

Now we check eQTL databases.

Result:

`text id="4jyxya"`

`rs39 → Gene A`

`rs40 → No eQTL`

rs41 → Gene B

Immediately:

text id="4l6ylz" Gene A Gene B

become strong candidates.

0.4.125 Why Fine-Mapping First?

Imagine starting with:

text id="z6qxxd" 1000 GWAS SNPs

Checking all eQTLs would produce enormous numbers of candidate genes.

Fine-mapping reduces the search space:

text id="l7v0pq" 1000 SNPs ↓ 5 SNPs

making interpretation much easier.

0.4.126 The Problem of Nearby Genes

A common beginner mistake is:

text id="fpf8gb" Nearest gene = causal gene

This is often incorrect.

Example:

text id="tvvrfw" GWAS SNP

may lie near:

text id="ebg4ja" Gene A

but actually regulate:

text id="hzl9mk" Gene B

located much farther away.

eQTL data help resolve this ambiguity.

0.4.127 Fine-Mapping eQTL Signals

Interestingly, eQTL studies themselves can be fine-mapped.

Workflow:

```

text id="ln09rl" Expression Trait          ↓ eQTL Mapping          ↓
Fine-Mapping          ↓ eQTL Credible Sets
  
```

This becomes important later when we discuss colocalization.

0.4.128 Example: GWAS and eQTL Fine-Mapping

Suppose:

GWAS fine-mapping finds:

```
text id="vcztuw" CS1 = {rs39}
```

and eQTL fine-mapping finds:

```
text id="0yxz6r" CS1 = {rs39}
```

This is highly suggestive.

Both studies point to:

```
text id="yxq9za" rs39
```

as the causal variant.

0.4.129 Biological Interpretation

Suppose:

```
text id="yz67gf" rs39
```

increases expression of:

```
text id="z7l5e8" Gene A
```

and also increases disease risk.

We may hypothesize:

```

text id="h0axc8" Higher Gene A Expression          ↓ Higher Disease Risk
  
```

This provides a mechanistic explanation.

0.4.130 Tissue Specificity

One of the most important aspects of eQTL analysis is:

0.4.130.1 Tissue Specificity

A variant may be an eQTL in:

```
text id="y8pf7v" Brain
```

but not in:

```
text id="1nsz8d" Blood
```

or

```
text id="qfxr9j" Liver
```

0.4.131 Why Tissue Matters

Suppose we study:

```
text id="k2f2p4" Schizophrenia
```

Relevant tissues might include:

```
```text id="9s4fdh" Prefrontal Cortex
```

Hippocampus

Neurons

```
Brain eQTLs are often more informative than blood eQTLs.
```

```

```

```
Example: Psychiatric Genetics
```

```
Workflow:
```

```
```text id="srlhdi"
```

```
Schizophrenia GWAS
```

```
↓
```

```
Fine-Mapping
```

```
↓
```

```
PsychENCODE eQTLs
```



Candidate Gene

This has become standard practice in psychiatric genomics.

0.4.132 What Fine-Mapping Alone Cannot Tell Us

Fine-mapping identifies:

text id="s1elxv" Likely causal variants

but not necessarily:

text id="n1mbjo" Target genes

eQTL analysis identifies:

text id="f1r91m" Gene regulation

but not necessarily:

text id="3gtx85" Disease causality

Combining both provides much stronger evidence.

0.4.133 Fine-Mapping + eQTL = Gene Prioritization

Together:

text id="h9zttx" Fine-Mapping

answers:

text id="ig5uv0" Which SNP?

while:

text id="uknxxl" eQTL Analysis

answers:

text id="dtm8lv" Which Gene?

0.4.134 Remaining Problem

Even if:

`text id="sgw6gs" GWAS SNP`

and

`text id="hnp7lu" eQTL SNP`

appear similar,

we still cannot conclude:

`text id="aq5u80" Same causal variant`

because LD can create misleading overlap.

This is the exact motivation for:

0.4.134.1 Colocalization

which formally tests whether two association signals are driven by the same causal variant.

This is the subject of the next section.

0.4.135 Example End-to-End Workflow

`text id="agkt65" GWAS ↓ Fine-Mapping ↓ Credible Set ↓ eQTL
Lookup ↓ Candidate Gene ↓ Colocalization ↓ Shared Causal
Variant? ↓ Biological Mechanism`

This workflow is now standard in many GWAS publications.

0.4.136 Key Takeaways

Fine-mapping identifies likely causal variants, but it does not directly reveal which gene is affected. eQTL analysis provides the missing link by identifying variants that influence gene expression. By intersecting fine-mapped GWAS signals with eQTL results, researchers can prioritize candidate target genes and generate mechanistic hypotheses about disease biology. However, overlap between GWAS and eQTL signals does not automatically imply a shared causal variant because linkage disequilibrium can create misleading patterns. This limitation motivates the use of colocalization methods, which formally test whether two traits are driven by the same underlying variant.

0.4.137 Part 12: Colocalization — Determining Whether a GWAS Signal and an eQTL Signal Share the Same Causal Variant

In the previous chapter, we learned how to use eQTL data to identify candidate target genes for a GWAS locus.

However, a critical problem remains.

Suppose we observe:

GWAS Signal

and

eQTL Signal

in the same genomic region.

Does that mean:

Same Variant

causes both signals?

Unfortunately:

No.

Not necessarily.

This is exactly why colocalization methods were developed.

The ISG workshop describes colocalization as the formal statistical test for determining whether two association signals are driven by the same underlying causal variant.

0.4.138 Why We Need Colocalization

Suppose a schizophrenia GWAS identifies:

rs39

as a likely causal variant.

At the same locus, an eQTL study reports:

rs40

associated with expression of:

Gene A

Because:

rs39 ↔ rs40

are in strong LD:

$$r^2 = 0.95$$

both signals appear to overlap.

But two possibilities exist.

0.4.139 Scenario 1: Shared Causal Variant

rs39



Gene A Expression

rs39



Disease Risk

Same variant drives both traits.

This is true colocalization.

0.4.140 Scenario 2: Distinct Variants

rs39



Disease Risk

and

rs40

↓

Gene A Expression

Different variants.

Strong LD merely makes them appear similar.

This is NOT colocalization.

0.4.141 The Goal of Colocalization

Colocalization asks:

Do both traits
share the same
causal variant?

This is one of the most important questions in modern statistical genetics.

0.4.142 The Five Colocalization Hypotheses

The coloc framework evaluates five mutually exclusive hypotheses.

0.4.143 H0

No association
with either trait

Diagram:

Trait 1 x

Trait 2 x

Nothing is happening.

0.4.144 H1

Association
with Trait 1 only

Diagram:

Trait 1 ✓

Trait 2 ✗

Only the GWAS signal exists.

0.4.145 H2

Association
with Trait 2 only

Diagram:

Trait 1 ✗

Trait 2 ✓

Only the eQTL signal exists.

0.4.146 H3

Both traits associated

Different causal variants

Diagram:

Variant A

↓

Trait 1

Variant B

↓

Trait 2

This is often called:

Linkage

or

Distinct Signals

0.4.147 H4

Both traits associated

One shared
causal variant

Diagram:

Variant A



Trait 1

Variant A



Trait 2

This is the outcome most researchers hope to see.

0.4.148 Posterior Probabilities

coloc computes:

PP.H0

PP.H1

PP.H2

PP.H3

PP.H4

These probabilities sum to:

1

or

100%

0.4.149 Example

Suppose coloc reports:

$PP.H0 = 0.00$

$PP.H1 = 0.00$

$PP.H2 = 0.01$

$PP.H3 = 0.05$

$PP.H4 = 0.94$

Interpretation:

94% probability
of a shared
causal variant

Strong evidence for colocalization.

0.4.150 Interpreting PP.H4

Typical interpretation:

$PP.H4 > 0.80$

Strong evidence.

$PP.H4 > 0.90$

Very strong evidence.

$$PP.H4 > 0.95$$

Extremely strong evidence.

0.4.151 Interpreting PP.H3

Suppose:

$$PP.H3 = 0.95$$

Interpretation:

Both traits
are associated

BUT

through different variants

This is a very different biological conclusion.

0.4.152 coloc.abf

The original coloc method is:

0.4.152.1 coloc.abf

The ISG workshop describes it as a summary-statistics method requiring no LD matrix.

0.4.153 Inputs

Beta

SE

Sample Size

for both traits.

Example:

GWAS Summary Statistics

+

eQTL Summary Statistics

0.4.154 Major Assumption

coloc.abf assumes:

One causal variant
per locus

Exactly the same limitation as ABF fine-mapping.

0.4.155 Why This Is a Problem

Real loci often contain:

Signal 1

Signal 2

Signal 3

The workshop explicitly simulates a locus with multiple causal variants.

In such situations:

coloc.abf

can miss true colocalization.

0.4.156 Example from the Workshop

Trait 1 (GWAS):

Causal Variant 39

Trait 2 (eQTL):

Variant 39
(shared)

Variant 20
(private)

The eQTL contains two signals.

1 Result

coloc.abf strongly favored:

H3

meaning:

Different variants

even though a shared variant truly existed.

This demonstrates the limitation of the single-causal-variant assumption.

1.0.1 coloc.susie

To solve this problem:

coloc.susie

was developed.

The ISG workshop recommends it whenever reliable LD information is available.

1.0.2 Main Idea

Instead of comparing:

One signal

against

One signal

it compares:

Credible Set 1

vs

Credible Set 1

Credible Set 2

vs

Credible Set 2

and so on.

1.0.3 Workflow

GWAS



SuSiE



Credible Sets

eQTL



SuSiE



Credible Sets



coloc.susie

1.0.4 Example Output

The workshop reports results like:

GWAS Signal 39

eQTL Signal 39

$$PP.H4 = 1.0$$

meaning essentially perfect evidence for colocalization.

1.0.5 Why coloc.susie Is Preferred

Advantages:

Multiple causal variants

Credible-set aware

Handles complex loci

Uses fine-mapping results

This makes it the modern standard.

1.0.6 Biological Interpretation

Suppose:

GWAS

colocalizes with

Brain eQTL

for:

CACNA1C

with:

$$PP.H4 = 0.97$$

Interpretation:

Strong evidence

that the same variant

influences both

CACNA1C expression

and disease risk

This creates a compelling mechanistic hypothesis.

1.0.7 What Colocalization Does NOT Prove

Even a high:

PP.H4

does NOT prove:

Causality

It means:

The data are consistent
with one shared variant

Additional biological validation is still required.

1.0.8 Key Takeaways

Colocalization is the statistical framework used to determine whether two association signals are driven by the same causal variant. The coloc framework evaluates five hypotheses (H_0 – H_4), with H_4 representing a shared causal variant and H_3 representing distinct causal variants. The original coloc.abf method requires only summary statistics but assumes a single causal variant per locus. Modern analyses increasingly use coloc.susie because it leverages SuSiE fine-mapping results and can accommodate multiple causal variants. Colocalization serves as the critical bridge between GWAS findings and gene regulatory mechanisms, helping researchers connect disease-associated variants to specific genes and biological pathways.

1.0.9 Part 13: Practical Interpretation of Colocalization Results — Reading PP.H0–PP.H4, Common Pitfalls, and Real-World Applications

Running a colocalization analysis is relatively easy.

Interpreting the results correctly is much harder.

Many beginners see:

```
PP.H4 = 0.85
```

and immediately conclude:

```
Gene identified!
```

Unfortunately, interpretation requires considerably more care.

In this chapter we will learn:

- How to interpret PP.H0–PP.H4
- What constitutes strong evidence
- Common mistakes
- Sensitivity analyses
- Real-world examples
- How fine-mapping quality affects colocalization

This is the final skill needed before applying colocalization in real studies.

1.0.10 The Most Important Output

After running:

```
coloc.abf(...)
```

or

```
coloc.susie(...)
```

you typically obtain:

```
PP.H0  
PP.H1
```

PP.H2

PP.H3

PP.H4

These probabilities always sum to:

1

1.0.11 Interpretation Refresher

1.0.11.1 H0

No association
with either trait

1.0.11.2 H1

Trait 1 only

Example:

GWAS signal exists

eQTL signal absent

1.0.11.3 H2

Trait 2 only

Example:

eQTL signal exists

GWAS signal absent

1.0.11.4 H3

Both traits associated

Different causal variants

1.0.11.5 H4

Both traits associated

Shared causal variant

This is usually the desired outcome.

1.0.12 Example 1: Strong Colocalization

Suppose results are:

$$PP.H0 = 0.00$$

$$PP.H1 = 0.01$$

$$PP.H2 = 0.02$$

$$PP.H3 = 0.03$$

$$PP.H4 = 0.94$$

Interpretation:

Very strong evidence
for a shared causal variant

This is the textbook example of successful colocalization.

1.0.13 Example 2: Strong Evidence Against Colocalization

Suppose:

$$PP.H0 = 0.00$$

$$PP.H1 = 0.00$$

$$PP.H2 = 0.00$$

$$PP.H3 = 0.96$$

$$PP.H4 = 0.04$$

Interpretation:

Both traits are associated

BUT

different causal variants
appear responsible

This is one of the most common outcomes.

1.0.14 Example 3: Ambiguous Region

Suppose:

$$PP.H0 = 0.05$$

$$PP.H1 = 0.10$$

$$PP.H2 = 0.10$$

$$PP.H3 = 0.35$$

$$PP.H4 = 0.40$$

Interpretation:

Data are inconclusive

Neither:

H3

nor

H4

dominates.

Additional data may be needed.

1.0.15 Practical Thresholds

Many studies use:

1.0.15.1 Weak Evidence

$$PP.H4 < 0.50$$

1.0.15.2 Moderate Evidence

$$0.50 \leq PP.H4 < 0.80$$

1.0.15.3 Strong Evidence

$$PP.H4 \geq 0.80$$

1.0.15.4 Very Strong Evidence

$$PP.H4 \geq 0.90$$

1.0.15.5 Exceptional Evidence

$$PP.H4 \geq 0.95$$

These thresholds are conventions rather than strict rules.

Context always matters.

1.0.16 Looking Beyond PP.H4

Many beginners focus only on:

$$PP.H4$$

This is dangerous.

Consider:

$$PP.H3 = 0.45$$

$$PP.H4 = 0.50$$

Although:

$$PP.H4$$

is the largest probability,

there remains substantial support for:

Different causal variants

The result is uncertain.

1.0.17 The H3 Versus H4 Competition

In practice:

H3

and

H4

are often the most important hypotheses.

The real question becomes:

Shared Variant?

or

Different Variants?

A useful diagnostic is:

$$PP.H4 / (PP.H3 + PP.H4)$$

Large values indicate stronger evidence for sharing.

1.0.18 Example

Suppose:

$$PP.H3 = 0.10$$

$$PP.H4 = 0.90$$

Then:

$$0.90 / (0.90 + 0.10)$$

$$= 0.90$$

Excellent support for sharing.

1.0.19 Why Fine-Mapping Quality Matters

Colocalization depends on fine-mapping.

Poor fine-mapping leads to poor colocalization.

Example:

GWAS Credible Set

=

50 SNPs

and

eQTL Credible Set

=

60 SNPs

The uncertainty becomes enormous.

Colocalization may become inconclusive.

1.0.20 High-Resolution Example

GWAS:

{rs39}

eQTL:

{rs39}

Result:

$PP.H4 \approx 1$

Much easier to interpret.

1.0.21 Common Mistake #1

1.0.21.1 Assuming Overlap Equals Colocalization

Researchers sometimes observe:

GWAS peak

and

eQTL peak

in the same region.

They conclude:

Same signal

This is incorrect.

Strong LD can create apparent overlap even when different variants are responsible.

This is exactly why formal colocalization methods exist.

1.0.22 Common Mistake #2

1.0.22.1 Ignoring LD Mismatch

Suppose:

GWAS

uses:

European samples

while:

eQTL

uses:

African samples

LD patterns differ.

Fine-mapping may identify different credible sets even when the biology is identical.

Always consider ancestry.

1.0.23 Common Mistake #3

1.0.23.1 Assuming Colocalization Proves Causality

Even:

$PP.H4 = 0.99$

does not prove:

Gene A causes disease.

It only suggests:

Shared genetic regulation.

Further evidence may be required:

- Functional studies
 - CRISPR experiments
 - Animal models
 - Perturbation experiments
-

1.0.24 Common Mistake #4

1.0.24.1 Ignoring Tissue Context

Suppose:

Brain eQTL

$PP.H4 = 0.95$

and

Blood eQTL

$PP.H4 = 0.05$

for schizophrenia.

The brain result is biologically more relevant.

Tissue selection matters enormously.

1.0.25 Sensitivity Analysis

A good colocalization study should evaluate:

Different priors

because Bayesian results depend on prior assumptions.

The coloc package allows prior specification.

Example:

```
coloc.abf(  
  dataset1,  
  dataset2,  
  p1 = 1e-4,  
  p2 = 1e-4,  
  p12 = 1e-5  
)
```

Researchers often test multiple prior settings.

1.0.26 What Are Priors?

Priors represent beliefs about:

How likely a SNP is
to affect Trait 1

How likely a SNP is
to affect Trait 2

How likely a SNP affects both

Different assumptions can influence results.

1.0.27 Example of Sensitivity

Suppose:

$PP.H4 = 0.82$

under one prior.

and

$PP.H4 = 0.35$

under another.

Interpretation:

Result is unstable

Be cautious.

1.0.28 Real-World Example Workflow

Consider:

Depression GWAS

and

Brain eQTL Data

Workflow:

```
graph TD
  GWAS --> FM1[Fine-Mapping]
  FM1 --> CS1[Credible Sets]
  CS1 --> BEQTL[Brain eQTL]
  BEQTL --> FM2[Fine-Mapping]
  FM2 --> CS2[Credible Sets]
  CS2 --> coloc[coloc.susie]
```

Result:

PP.H4 = 0.92

Interpretation:

Strong evidence
that the same variant
influences both

gene expression

and depression risk

This provides a biologically meaningful hypothesis.

1.0.29 What Should Be Reported?

A publication should typically report:

Gene

Tissue

PP.H4

PP.H3

Method Used

Credible Set Sizes

Fine-Mapping Method

Example:

Gene:

CACNA1C

Tissue:

Prefrontal Cortex

Method:

coloc.susie

PP.H4:

0.96

GWAS CS Size:

2

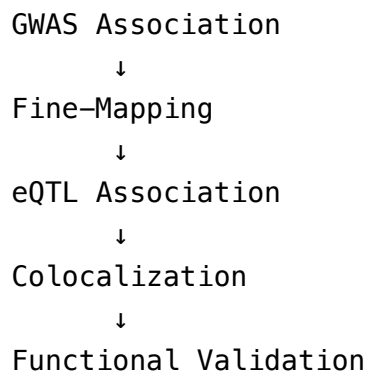
eQTL CS Size:

1

This provides a complete picture.

1.0.30 The Hierarchy of Evidence

Increasing confidence:



Each step strengthens the evidence.

1.0.31 Practical Checklist

Before trusting a colocalization result, ask:

1.0.31.1 Are both traits associated?

Check H1/H2/H3/H4

1.0.31.2 Is PP.H4 high?

Preferably > 0.8

1.0.31.3 Is PP.H3 low?

Different-variant explanation
should be weak

1.0.31.4 Were credible sets small?

Better resolution

1.0.31.5 Is LD appropriately matched?

Critical

1.0.31.6 Is the tissue biologically relevant?

Very important

1.0.32 Were sensitivity analyses performed?

Highly recommended

1.0.33 Key Takeaways

Colocalization results should be interpreted using the full set of posterior probabilities rather than PP.H4 alone. Strong evidence for colocalization typically requires a high PP.H4 and a low PP.H3. Fine-mapping quality directly affects colocalization performance because poorly resolved credible sets introduce uncertainty. Colocalization does not prove causality but provides strong evidence that the same genetic variant influences both a molecular trait and a disease trait. Proper interpretation requires attention to ancestry matching, tissue relevance, prior assumptions, and sensitivity analyses.

1.0.34 Part 14: End-to-End Statistical Genetics Pipeline — From GWAS to Biological Discovery

We have now reached the final chapter of this tutorial.

Throughout the previous chapters, we explored:

- GWAS
- Fine-Mapping
- eQTL Analysis
- Colocalization

individually.

In practice, however, researchers do not perform these analyses in isolation.

Instead, they are combined into a single integrated workflow whose ultimate goal is:

Move from

Association

to

Biological Mechanism

This chapter ties together everything we have learned into one coherent pipeline.

1.0.35 The Fundamental Problem

Suppose a GWAS identifies:

Chromosome 6

as associated with schizophrenia.

GWAS tells us:

Something important
exists in this region.

But it does not tell us:

Which variant?

Which gene?

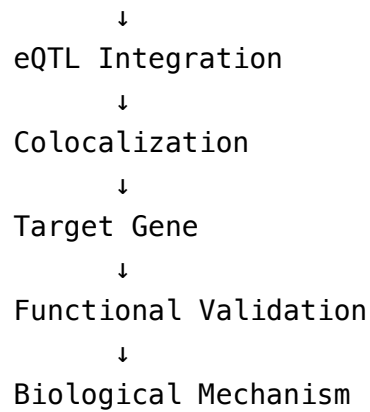
Which mechanism?

This is the motivation behind the modern post-GWAS pipeline.

1.0.36 The Complete Workflow

The modern statistical genetics workflow can be summarized as:

```
Genotypes
  ↓
GWAS
  ↓
Significant Locus
  ↓
Fine-Mapping
  ↓
Credible Sets
```



This framework underlies many contemporary studies from:

- UK Biobank
- FinnGen
- Psychiatric Genomics Consortium
- GTEx
- PsychENCODE

1.0.37 Step 1: GWAS

Everything begins with a Genome-Wide Association Study.

Input:

Genotypes

Phenotypes

Covariates

Example:

500,000 individuals

10 million SNPs

Typical GWAS model:

Phenotype

=

SNP

+

Covariates

+

Error

Output:

Beta

SE

P-value

for every SNP.

1.0.38 Example GWAS Result

Suppose:

rs39

 $p = 1 \times 10^{-12}$

This indicates:

Strong association

but not causality.

1.0.39 Limitation of GWAS

Because of LD:

rs39

rs40

rs41

rs42

may all appear significant.

GWAS cannot determine:

Which SNP
is causal.

This leads to the next step.

1.0.40 Step 2: Fine-Mapping

Fine-mapping attempts to identify:

Likely causal variants

rather than merely associated variants.

Inputs:

GWAS Summary Statistics

+

LD Matrix

Methods:

SuSiE

FINEMAP

CARMA

ABF

1.0.41 Fine-Mapping Output

Example:

rs39

PIP = 0.94

and

95% Credible Set

=

{rs39, rs40}

Interpretation:

The causal variant
is probably one of these SNPs.

1.0.42 Why Fine-Mapping Matters

Without fine-mapping:

100 SNPs

may require investigation.

After fine-mapping:

2 SNPs

remain.

This dramatically improves biological interpretation.

1.0.43 Step 3: Gene Prioritization

Now we ask:

Which gene
is affected?

The nearest gene is often:

Not
the causal gene.

Therefore we integrate functional genomics.

1.0.44 Step 4: eQTL Analysis

An eQTL identifies variants affecting gene expression.

Example:

rs39

influences:

CACNA1C expression

in brain tissue.

This creates a hypothesis:

rs39

↓

CACNA1C Expression

↓

Disease Risk

1.0.45 Why eQTLs Are Useful

Fine-mapping answers:

Which SNP?

eQTL analysis answers:

Which gene?

Combining them dramatically improves interpretation.

1.0.46 Step 5: Fine-Mapping the eQTL

Modern analyses often fine-map:

GWAS

and

eQTL

independently.

Example:

GWAS:

$CS = \{rs39\}$

eQTL:

$CS = \{rs39\}$

This is highly suggestive.

But still not proof.

1.0.47 Step 6: Colocalization

Now we ask:

Do both traits
share the same
causal variant?

This is the role of:

`coloc.abf`

`coloc.susie`

1.0.48 The Five Hypotheses

Colocalization evaluates:

H_0

No association

H1

Trait 1 only

H2

Trait 2 only

H3

Different variants

H4

Shared variant

1.0.49 Example

Suppose:

$$PP.H4 = 0.96$$

Interpretation:

Strong evidence
for a shared
causal variant.

This is often the key result in modern post-GWAS studies.

1.0.50 Putting It Together

Suppose:

GWAS finds:

rs39

Fine-mapping finds:

$$PIP = 0.98$$

eQTL analysis finds:

rs39 affects CACNA1C

Colocalization finds:

PP.H4 = 0.97

This supports:

rs39
↓
CACNA1C Expression
↓
Schizophrenia Risk

A biologically meaningful hypothesis has emerged.

1.0.51 What Have We Achieved?

Originally:

10 million SNPs

were tested.

Now we have:

1 variant

1 gene

1 mechanism

This is the power of the integrated pipeline.

1.0.52 Real Example: Psychiatric Genetics

A common workflow in psychiatric genomics:

Schizophrenia GWAS



Fine-Mapping



Brain eQTL Data



Colocalization



Candidate Gene



Functional Validation

Resources often include:

PsychENCODE

GTEx

CommonMind

1.0.53 Beyond eQTLs

Modern studies increasingly integrate:

sQTLs

(splicing QTLs)

pQTLs

(protein QTLs)

mQTLs

(methylation QTLs)

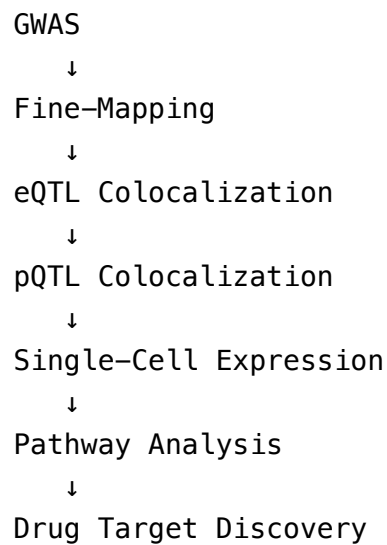
caQTLs

(chromatin accessibility QTLs)

The same fine-mapping and colocalization framework applies.

1.0.54 Multi-Omics Integration

A modern workflow may look like:



This represents the current frontier of statistical genetics.

1.0.55 Where Single-Cell Data Fits

Recently, many studies have added:

Single-Cell RNA-seq

to identify:

Which cell types
express the target gene?

Example:

Gene A

may be highly expressed in:

Excitatory Neurons

but not:

Microglia

providing additional biological context.

1.0.56 Where TWAS Fits

Another increasingly popular method is:

TWAS

Workflow:

GWAS
↓
Predicted Expression
↓
Gene-Level Association

TWAS complements:

Fine-Mapping

eQTL Analysis

Colocalization

and is often used together with them.

1.0.57 Where Mendelian Randomization Fits

Another downstream step is:

Mendelian Randomization

Workflow:

eQTL
↓
Gene Expression
↓
Disease

MR can test:

Is altered expression
causally related
to disease?

This provides stronger evidence than colocalization alone.

1.0.58 The Hierarchy of Evidence

Increasing confidence:

GWAS
↓
Fine-Mapping
↓
eQTL Association
↓
Colocalization
↓
TWAS
↓
Mendelian Randomization
↓
Functional Validation

Each layer adds evidence.

1.0.59 The Ultimate Goal

The goal is not simply:

Find significant SNPs

The real goal is:

Understand biology

Specifically:

Variant
↓
Gene

↓
Cell Type
↓
Pathway
↓
Disease

1.0.60 Common Beginner Misconception

Many researchers stop after:

GWAS

However:

GWAS

is often just the starting point.

The real biological discoveries usually emerge during:

Fine-Mapping

eQTL Integration

Colocalization

Functional Interpretation

1.0.61 A Modern Statistical Geneticist's Workflow

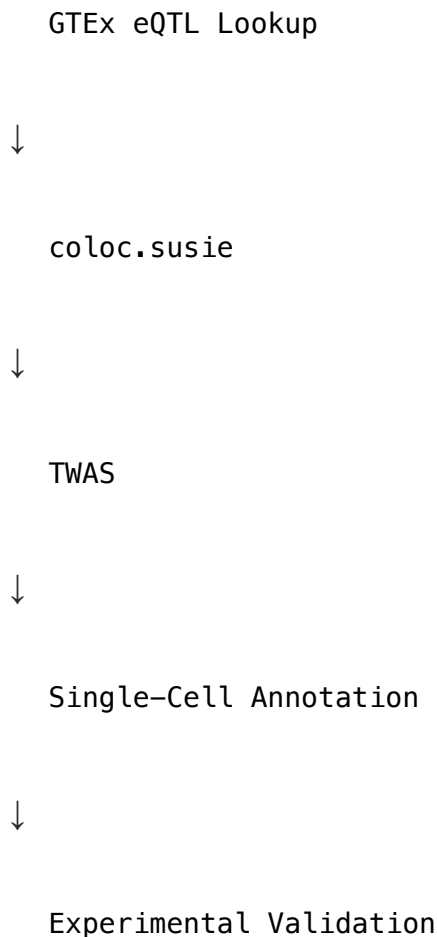
Today, a typical project might involve:

GWAS

↓

SuSiE Fine-Mapping

↓



This represents the modern state of the field.

1.0.62 Final Key Takeaways

The journey from GWAS to biological discovery involves several interconnected analytical steps. GWAS identifies associated genomic regions, fine-mapping narrows these regions to likely causal variants, eQTL analysis links variants to gene expression, and colocalization tests whether the same variant influences both expression and disease risk. Together, these approaches transform statistical associations into biologically meaningful hypotheses. Modern statistical genetics increasingly integrates additional layers such as single-cell genomics, TWAS, Mendelian Randomization, and multi-omics datasets to build a comprehensive understanding of disease mechanisms. The ultimate objective is to move from a significant SNP to a causal gene, a relevant cell type, a biological pathway, and eventually therapeutic insight.
